
Analysis of Muon with Nesterov Momentum

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Abstract

1 0.5 page

2 1 Introduction

3 [2 pages] We are interested in solving the optimization problem

$$\min_{\mathbf{X} \in \mathbb{R}^{d \times d}} f(\mathbf{X})$$

4 Here we consider the batched Muon with Nesterov’s momentum and exact Polar decomposition, i.e.

$$\begin{aligned} \mathbf{G}_k &= \sum_{i=1}^B \mathbf{G}_k^i / B \\ \mathbf{C}_k &= \gamma_k \mathbf{C}_{k-1} + \mathbf{G}_k \\ \mathbf{M}_k &= \beta_k \mathbf{C}_k + \mathbf{G}_k \\ \mathbf{X}_{k+1} &= \mathbf{X}_k - \eta_k \mathbf{O}(\mathbf{M}_k) \end{aligned} \tag{1}$$

5 for some step size $\eta_k > 0$ and momentum parameters $\gamma_k, \beta_k \in (0, 1)$. In the analysis, we will choose
6 $\gamma_k = \beta_{k-1}$ or $\gamma_k = \beta_k$. Moreover, to analyze the heavy-tailed noise distribution, we consider the
7 following assumption on $\mathbf{G}_k^i$

8 The update rule of the Muon with Nesterov’s momentum is given by,

$$\begin{aligned} \mathbf{G}_k &= \sum_{i=1}^B \mathbf{G}_k^i / B \\ \mathbf{C}_k &= \beta \mathbf{C}_{k-1} + \mathbf{G}_k \\ \mathbf{M}_k &= \beta \mathbf{C}_k + \mathbf{G}_k \\ \mathbf{X}_{k+1} &= \mathbf{X}_k - \eta T(\mathbf{M}_k) \end{aligned}$$

9 where $\beta, \eta > 0$ are the momentum and step size, respectively, and $\mathbf{G}_k$ is an estimate of the gradient
10 $\nabla f(\mathbf{X}_k)$, and $T(\mathbf{M}_k)$ is the estimate of the polar decomposition of $\mathbf{M}_k$. Note that we can expand
11 Nesterov’s momentum as follows.

$$\begin{aligned} \mathbf{M}_k &= \beta \mathbf{C}_k + \mathbf{G}_k \\ &= \beta(\beta \mathbf{C}_{k-1} + \mathbf{G}_k) + \mathbf{G}_k \\ &= \beta(\mathbf{M}_{k-1} - \mathbf{G}_{k-1} + \mathbf{G}_k) + \mathbf{G}_k \\ &= \beta \mathbf{M}_{k-1} + \mathbf{G}_k + \beta(\mathbf{G}_k - \mathbf{G}_{k-1}). \end{aligned}$$

12 1.1 Main Contributions

13 We summarize the main contributions of the paper below.

- 14 • First Analysis of Muon with Nesterov’s Momentum (to the best of our knowledge).
- 15 • Analysis under Relaxed Assumptions on the Stochastic Gradient.

- 16 • Optimal Sample Complexity.
- 17 • Convergence without the knowledge of α .
- 18 • Novelty in Proof Techniques
- 19 • Analysis for Inexact Polar Decomposition. Our assumption covers the Newton Schulz
- 20 iteration as a special case.
- 21 • Numerical Experiments to demonstrate the performance of Nesterov's momentum.

22 **2 Related Work**

23 1 page

- 24 • Introduction of spectral GD, Muon, and related algorithms.
- 25 • Discussion on the analysis of the muon algorithm.
- 26 • Discussion on the optimal sample complexity for non-convex functions.
- 27 • Assumption on Stochastic Gradient like, bounded variance. Heavy tailed distributions.

28 3 Convergence Analysis

29 3.1 Assumptions

30 0.75 page

Assumption 3.1. f is L -smooth i.e.

$$f(\mathbf{Y}) \leq f(\mathbf{X}) + \langle \nabla f(\mathbf{X}), \mathbf{Y} - \mathbf{X} \rangle + \frac{L}{2} \|\mathbf{X} - \mathbf{Y}\|_F^2.$$

Assumption 3.2 (Generalized Heavy-Tailed Noises). $\mathbf{G}_k^i$ is an unbiased estimate of $\nabla f(\mathbf{X}_k)$ i.e. $\mathbb{E}[\mathbf{G}_k^i] = \nabla f(\mathbf{X}_k)$ and satisfies

$$\mathbb{E}[\|\mathbf{G}_k^i - \nabla f(\mathbf{X}_k)\|_F^\alpha | \mathbf{X}_k] \leq \sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_k)\|_F^\alpha$$

with $\alpha \in (1, 2]$ for all $i \in [B]$.

31 Note that we can use the above assumption to get a bound on the noise level of $\mathbf{G}_k$ as fol-
32 lows.

Lemma 3.3. Under Assumption 3.2, we have

$$\mathbb{E}[\|\mathbf{G}_k - \nabla f(\mathbf{X}_k)\|_F^\alpha | \mathbf{X}_k] \leq \frac{C_\alpha}{B^{\alpha-1}} (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_k)\|_F^\alpha) \quad (2)$$

for some $C_\alpha > 0$ independent of batch size B .

33 *Proof.* From the definition of $\mathbf{G}_k := \frac{1}{B} \sum_{i=1}^B \mathbf{G}_k^i$, we have

$$\begin{aligned} \mathbb{E}[\|\mathbf{G}_k - \nabla f(\mathbf{X}_k)\|_F^\alpha | \mathbf{X}_k] &= \mathbb{E}\left[\left\|\frac{1}{B} \sum_{i=1}^B \mathbf{G}_k^i - \frac{1}{B} \sum_{i=1}^B \nabla f(\mathbf{X}_k)\right\|_F^\alpha | \mathbf{X}_k\right] \\ &= \frac{1}{B^\alpha} \mathbb{E}\left[\left\|\sum_{i=1}^B (\mathbf{G}_k^i - \nabla f(\mathbf{X}_k))\right\|_F^\alpha | \mathbf{X}_k\right] \\ &\leq \frac{C_\alpha}{B^\alpha} \sum_{i=1}^B \mathbb{E}[\|\mathbf{G}_k^i - \nabla f(\mathbf{X}_k)\|_F^\alpha | \mathbf{X}_k] \end{aligned}$$

34 where the last inequality follows from the Bahr-Essen inequality [Wu and Wang, 2023] for some
35 constant $C_\alpha > 0$ independent of batch size B (as $\mathbf{G}_k^i - \nabla f(\mathbf{X}_k)$ are mean zero i.i.d. random
36 variables conditioned on $\mathbf{X}_k$). Then, using Assumption 3.2, we have

$$\begin{aligned} \mathbb{E}[\|\mathbf{G}_k - \nabla f(\mathbf{X}_k)\|_F^\alpha | \mathbf{X}_k] &\leq \frac{C_\alpha}{B^\alpha} \sum_{i=1}^B (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_k)\|_F^\alpha) \\ &= \frac{C_\alpha}{B^{\alpha-1}} (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_k)\|_F^\alpha). \end{aligned}$$

37 Thus, the noise of $\mathbf{G}_k$ decreases as batch size B increases. $\square$

38 3.2 Analysis with the Knowledge of α

39 [0.75 page] For our analysis, we define rescaled momentum variables

$$\widetilde{\mathbf{C}}_k := (1 - \beta) \mathbf{C}_k, \quad (3)$$

$$\widetilde{\mathbf{M}}_k := (1 - \beta) \mathbf{M}_k. \quad (4)$$

40 Note that for this rescaled momentum, we have

$$O(\mathbf{M}_k) = O(\widetilde{\mathbf{M}}_k),$$

41 as scaling only changes the singular values, while the singular vectors remain unchanged. Moreover,
42 we can show that $\widetilde{\mathbf{M}}_k$ satisfies the following recursion.

Lemma 3.4. For (3) and (4), we have

$$\widetilde{\mathbf{M}}_k = \beta \widetilde{\mathbf{M}}_{k-1} + (1 - \beta)((1 + \beta)\mathbf{G}_k - \beta \mathbf{G}_{k-1}).$$

43 *Proof.* From (3), we obtain

$$\begin{aligned} \widetilde{\mathbf{C}}_k &= (1 - \beta)\mathbf{C}_k \\ &= (1 - \beta)(\beta \mathbf{C}_{k-1} + \mathbf{G}_k) \\ &= \beta \widetilde{\mathbf{C}}_{k-1} + (1 - \beta)\mathbf{G}_k, \end{aligned}$$

44 and from (4), we get

$$\begin{aligned} \widetilde{\mathbf{M}}_k &= (1 - \beta)\mathbf{M}_k \\ &= (1 - \beta)(\beta \mathbf{C}_k + \mathbf{G}_k) \\ &= \beta \widetilde{\mathbf{C}}_k + (1 - \beta)\mathbf{G}_k. \end{aligned}$$

45 For $k \geq 1$, we want to express $\widetilde{\mathbf{M}}_k$ in terms of $\widetilde{\mathbf{M}}_{k-1}$. Note that

$$\begin{aligned} \widetilde{\mathbf{M}}_k &= \beta \widetilde{\mathbf{C}}_k + (1 - \beta)\mathbf{G}_k \\ &= \beta(\beta \widetilde{\mathbf{C}}_{k-1} + (1 - \beta)\mathbf{G}_k) + (1 - \beta)\mathbf{G}_k \\ &= \beta^2 \widetilde{\mathbf{C}}_{k-1} + (1 - \beta)(1 + \beta)\mathbf{G}_k. \end{aligned}$$

46 Also $\widetilde{\mathbf{M}}_{k-1} = \beta \widetilde{\mathbf{C}}_{k-1} + (1 - \beta)\mathbf{G}_{k-1}$, so $\beta \widetilde{\mathbf{C}}_{k-1} = \widetilde{\mathbf{M}}_{k-1} - (1 - \beta)\mathbf{G}_{k-1}$ and therefore

$$\widetilde{\mathbf{M}}_k = \beta \widetilde{\mathbf{M}}_{k-1} + (1 - \beta)((1 + \beta)\mathbf{G}_k - \beta \mathbf{G}_{k-1}).$$

47

□

48 3.3 Analysis without the Knowledge of α

49 0.75 page

50 3.4 Analysis for Inexact Polar Decomposition

51 [1.5 pages]

52 Assumption for Inexact Polar Decomposition Analysis. Also include randomization.

Assumption 3.5.

$$\begin{aligned} \mathbb{E} [\langle \mathbf{M}_k, T(\mathbf{M}_k) \rangle \mid \mathbf{M}_k] &\geq (1 - \delta_k) \|\mathbf{M}_k\|_* \\ \mathbb{E} [\|T(\mathbf{M}_k)\|_{\text{op}}^2 \mid \mathbf{M}_k] &\leq (1 + \rho_k)^2 \end{aligned}$$

53 • Note that $T(\mathbf{M}_k) = O(\mathbf{M}_k)$ satisfies this assumption with $\delta_k = \rho_k = 0$.

54 • Note that $\|O(\mathbf{M}_k) - T(\mathbf{M}_k)\|_{\text{op}} \leq \varepsilon_k$ implies

$$\begin{aligned} \langle \mathbf{M}_k, T(\mathbf{M}_k) \rangle &= \langle \mathbf{M}_k, O(\mathbf{M}_k) \rangle - \langle \mathbf{M}_k, O(\mathbf{M}_k) - T(\mathbf{M}_k) \rangle \\ &\geq \langle \mathbf{M}_k, O(\mathbf{M}_k) \rangle - \|\mathbf{M}_k\|_* \|O(\mathbf{M}_k) - T(\mathbf{M}_k)\|_{\text{op}} \\ &\geq (1 - \varepsilon_k) \|\mathbf{M}_k\|_*, \end{aligned}$$

55 and

$$\|T(\mathbf{M}_k)\|_{\text{op}} \leq (1 + \varepsilon_k).$$

56 Consider the singular value decomposition of $\mathbf{M} \in \mathbb{R}^{m \times n}$

$$\begin{aligned}\mathbf{M} &= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top, \\ \mathbf{\Sigma} &= \text{diag}(\sigma_1, \dots, \sigma_r)\end{aligned}$$

57 with $\sigma_1 \geq \dots \geq \sigma_r > 0$, where $r = \text{rank}(\mathbf{M})$. Then, the polar decomposition of $\mathbf{M}$ is given by

$$\mathbf{O}(\mathbf{M}) := \mathbf{U}\mathbf{V}^\top.$$

58 We say that an approximation $T(\mathbf{M})$ satisfies the inexact polar assumptions if there exist $\delta(\mathbf{M}) \in$
59 $[0, 1]$ and $\rho(\mathbf{M}) \geq 0$ such that

$$\begin{aligned}\langle \mathbf{M}, T(\mathbf{M}) \rangle &\geq (1 - \delta(\mathbf{M})) \|\mathbf{M}\|_*, \\ \|T(\mathbf{M})\|_{\text{op}} &\leq 1 + \rho(\mathbf{M}).\end{aligned}$$

60 Here, we show that the approximation we obtain through a few iterations of Newton Schulz satisfies
61 this assumption.

62 **Newton–Schulz approximation.** Fix a scalar $\alpha \geq \|\mathbf{M}\|_{\text{op}}$ and define the normalized cubic
63 Newton–Schulz iterates by

$$\mathbf{X}_{t+1} = \frac{3}{2}\mathbf{X}_t - \frac{1}{2}\mathbf{X}_t\mathbf{X}_t^\top\mathbf{X}_t, \quad t = 0, 1, \dots, q-1.$$

64 where $\mathbf{X}_0 = \frac{\mathbf{M}}{\alpha}$. Now, set $T_q(\mathbf{M}) := \mathbf{X}_q$ and we will show $T_q(\mathbf{M})$ satisfies the assumptions.

Theorem 3.6. Define $s_i := \frac{\sigma_i}{\alpha} \in (0, 1], i = 1, \dots, r$ and let

$$f(x) := \frac{3}{2}x - \frac{1}{2}x^3.$$

Moreover, define the sequence of polynomials $p_0(x) := x$ and $p_{t+1}(x) := f(p_t(x))$. Then the Newton-Schulz iterate $T_q(\mathbf{M})$ admits the representation

$$T_q(\mathbf{M}) = \mathbf{U} \text{diag}(p_q(s_1), \dots, p_q(s_r)) \mathbf{V}^\top.$$

Consequently, $T_q(\mathbf{M})$ satisfies the assumptions with the parameters

$$\begin{aligned}\rho_q(\mathbf{M}) &= 0, \\ \delta_q(\mathbf{M}) &= 1 - \frac{\sum_{i=1}^r \sigma_i p_q(s_i)}{\sum_{i=1}^r \sigma_i}.\end{aligned}$$

Equivalently,

$$\begin{aligned}\langle \mathbf{M}, T_q(\mathbf{M}) \rangle &= (1 - \delta_q(\mathbf{M})) \|\mathbf{M}\|_*, \\ \|T_q(\mathbf{M})\|_{\text{op}} &\leq 1.\end{aligned}$$

Theorem 3.7. Let $C_{\delta, \rho} := (1 - \delta) + \sqrt{d}(1 + \rho)$. For $\mu := (1 - \delta) - C_{\delta, \rho} \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}}$, Muon with inexact polar decomposition satisfies

$$\begin{aligned}\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq \frac{f(\mathbf{X}_0) - f_\star}{\eta K \mu} + \frac{Ld(1 + \rho)^2}{2\mu} \eta + \frac{C_{\delta, \rho}}{\mu} \left[\frac{\mathbb{E} \|\epsilon_0\|_F}{K(1 - \beta)} \right. \\ &\quad + \frac{\beta^2}{1 - \beta} L \eta \sqrt{d}(1 + \rho) + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1 - \beta)^{\frac{\alpha-1}{\alpha}} \\ &\quad \left. + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \frac{\|\nabla f(\mathbf{X}_0)\|_F}{K} \right].\end{aligned}$$

65 **4 Numerical Experiments**

66 Around 1.5 to 2 pages

67 **Possible Experiments**

- 68 • Implement randomized NS with Nesterov Momentum to check it's performance compared
69 to full subspace NS.
- 70 • Performance of Muon based different batch sizes. Plot a curve of batch size on the x-axis
71 and loss/ accuracy on the y-axis for a fixed budget/ oracle calls.
- 72 • Check if the NS iterations satisfy the inexact polar decomposition assumption in experiments.
73 This should be true for theoretical ones (as we also show in theory). Also, check for empirical
74 NS iterations.
- 75 • Compare the performance of Muon for different inexact solvers like NS with cubic, quintic
76 (theoretical), quintic (empirical), and polar express.

77 **5 Conclusion**

78 **References**

- 79 Y. Wu and X. Wang. The von bahr-esseen type inequality under sub-linear expectations and applica-
80 tions. *arXiv preprint arXiv:2312.08781*, 2023.

Lemma A.1. Under Assumption 3.1, Muon satisfies

$$\eta \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F \leq (f(\mathbf{X}_0) - f_*) + \frac{Ld}{2} K \eta^2 + \eta(1 + \sqrt{d}) \sum_{k=0}^{K-1} \mathbb{E} \|\mathbf{S}_k\|_F. \quad (5)$$

82 *Proof.* Since $\mathbf{O}(\widetilde{\mathbf{M}}_k)$ is orthogonal, $\|\mathbf{O}(\widetilde{\mathbf{M}}_k)\|_F = \sqrt{d}$ and from the update rule

$$\|\mathbf{X}_{k+1} - \mathbf{X}_k\|_F = \eta \sqrt{d}.$$

83 By L -smoothness property of f , we obtain

$$\begin{aligned} f(\mathbf{X}_{k+1}) &\leq f(\mathbf{X}_k) + \langle \nabla f(\mathbf{X}_k), \mathbf{X}_{k+1} - \mathbf{X}_k \rangle + \frac{L}{2} \|\mathbf{X}_{k+1} - \mathbf{X}_k\|_F^2 \\ &= f(\mathbf{X}_k) - \eta \langle \nabla f(\mathbf{X}_k), \mathbf{O}(\widetilde{\mathbf{M}}_k) \rangle + \frac{Ld}{2} \eta^2 \\ &= f(\mathbf{X}_k) - \eta \langle \widetilde{\mathbf{M}}_k, \mathbf{O}(\widetilde{\mathbf{M}}_k) \rangle + \eta \langle \widetilde{\mathbf{M}}_k - \nabla f(\mathbf{X}_k), \mathbf{O}(\widetilde{\mathbf{M}}_k) \rangle + \frac{Ld}{2} \eta^2 \\ &= f(\mathbf{X}_k) - \eta \|\widetilde{\mathbf{M}}_k\|_* + \eta \langle \mathbf{S}_k, \mathbf{O}(\widetilde{\mathbf{M}}_k) \rangle + \frac{Ld}{2} \eta^2 \\ &\leq f(\mathbf{X}_k) - \eta \|\widetilde{\mathbf{M}}_k\|_F + \eta \|\mathbf{S}_k\|_F \|\mathbf{O}(\widetilde{\mathbf{M}}_k)\|_F + \frac{Ld}{2} \eta^2 \\ &= f(\mathbf{X}_k) - \eta \|\widetilde{\mathbf{M}}_k\|_F + \eta \sqrt{d} \|\mathbf{S}_k\|_F + \frac{Ld}{2} \eta^2. \end{aligned}$$

84 Using $\|\widetilde{\mathbf{M}}_k\|_F = \|\nabla f(\mathbf{X}_k) + \mathbf{S}_k\|_F \geq \|\nabla f(\mathbf{X}_k)\|_F - \|\mathbf{S}_k\|_F$ gives

$$f(\mathbf{X}_{k+1}) \leq f(\mathbf{X}_k) - \eta \|\nabla f(\mathbf{X}_k)\|_F + \eta(1 + \sqrt{d}) \|\mathbf{S}_k\|_F + \frac{Ld}{2} \eta^2.$$

85 Taking total expectation and summing from $k = 0$ to $K - 1$ yields

$$\eta \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F \leq (f(\mathbf{X}_0) - f_*) + \frac{Ld}{2} K \eta^2 + \eta(1 + \sqrt{d}) \sum_{k=0}^{K-1} \mathbb{E} \|\mathbf{S}_k\|_F.$$

86 This completes the proof of the Lemma. $\square$

Lemma A.2.

$$\mathbf{S}_k = \beta^k \boldsymbol{\epsilon}_0 + \beta^2 \sum_{s=1}^k \beta^{k-s} D_s + (1 - \beta) \left((1 + \beta) \boldsymbol{\xi}_k + \sum_{s=1}^{k-1} \beta^{k+1-s} \boldsymbol{\xi}_s - \beta^k \boldsymbol{\xi}_0 \right). \quad (6)$$

87 *Proof.* Subtract $\nabla f(\mathbf{X}_k)$ and use $\mathbf{G}_k = \nabla f(\mathbf{X}_k) + \boldsymbol{\xi}_k$ to obtain

$$\begin{aligned} \mathbf{S}_k &= \widetilde{\mathbf{M}}_k - \nabla f(\mathbf{X}_k) \\ &= \beta \widetilde{\mathbf{M}}_{k-1} + (1 - \beta) ((1 + \beta) \mathbf{G}_k - \beta \mathbf{G}_{k-1}) - \nabla f(\mathbf{X}_k) \\ &= \beta (\boldsymbol{\epsilon}_{k-1} + \nabla f(\mathbf{X}_{k-1})) + (1 - \beta) ((1 + \beta) \mathbf{G}_k - \beta \mathbf{G}_{k-1}) - \nabla f(\mathbf{X}_k) \\ &= \beta \boldsymbol{\epsilon}_{k-1} + (1 - \beta) ((1 + \beta) \boldsymbol{\xi}_k - \beta \boldsymbol{\xi}_{k-1}) + \beta \nabla f(\mathbf{X}_{k-1}) + (1 - \beta^2) \nabla f(\mathbf{X}_k) \\ &\quad - (\beta - \beta^2) \nabla f(\mathbf{X}_{k-1}) - \nabla f(\mathbf{X}_k) \\ &= \beta \boldsymbol{\epsilon}_{k-1} + (1 - \beta) ((1 + \beta) \boldsymbol{\xi}_k - \beta \boldsymbol{\xi}_{k-1}) + \beta^2 (\nabla f(\mathbf{X}_{k-1}) - \nabla f(\mathbf{X}_k)). \end{aligned}$$

88 Hence defining $D_k := \nabla f(\mathbf{X}_{k-1}) - \nabla f(\mathbf{X}_k)$ gives the clean recursion:

$$\mathbf{S}_k = \beta \boldsymbol{\epsilon}_{k-1} + \beta^2 D_k + (1 - \beta) ((1 + \beta) \boldsymbol{\xi}_k - \beta \boldsymbol{\xi}_{k-1}), \quad k \geq 1. \quad (7)$$

89 Unrolling equation 7 yields (by induction):

$$\mathbf{S}_k = \beta^k \boldsymbol{\epsilon}_0 + \beta^2 \sum_{s=1}^k \beta^{k-s} D_s + (1 - \beta) \left((1 + \beta) \boldsymbol{\xi}_k + \sum_{s=1}^{k-1} \beta^{k+1-s} \boldsymbol{\xi}_s - \beta^k \boldsymbol{\xi}_0 \right). \quad (8)$$

90 $\square$

Lemma A.3.

$$\begin{aligned} \mathbb{E}\|\mathbf{S}_k\|_F &\leq \beta^k \mathbb{E}\|\boldsymbol{\epsilon}_0\|_F + \frac{\beta^2}{1-\beta} L\eta\sqrt{d} + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} \\ &\quad + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} (1-\beta) \left((1+\beta) \mathbb{E}\|\nabla f(\mathbf{X}_k)\|_F + \sum_{s=1}^{k-1} \beta^{k+1-s} \mathbb{E}\|\nabla f(\mathbf{X}_s)\|_F + \beta^k \|\nabla f(\mathbf{X}_0)\|_F \right). \end{aligned} \quad (9)$$

91 *Proof.* By L -smoothness,

$$\|D_s\|_F = \|\nabla f(\mathbf{X}_{s-1}) - \nabla f(\mathbf{X}_s)\|_F \leq L\|\mathbf{X}_s - \mathbf{X}_{s-1}\|_F = L\eta\sqrt{d}.$$

92 Hence from equation ??,

$$\beta^2 \sum_{s=1}^k \beta^{k-s} \mathbb{E}\|D_s\|_F \leq \beta^2 L\eta\sqrt{d} \sum_{j=0}^{k-1} \beta^j \leq \frac{\beta^2}{1-\beta} L\eta\sqrt{d}.$$

93 View $\mathbb{R}^{d \times d}$ as $\mathbb{R}^{d^2}$ with the Frobenius norm. Because $\mathbb{E}[\boldsymbol{\xi}_s \mid \mathcal{F}_{s-1}] = 0$, the sequence $\{\boldsymbol{\xi}_s\}$ is a
94 martingale difference sequence.

95 Let the coefficients for the noise combination in equation 6 be defined as:

$$a_0 := -(1-\beta)\beta^k, \quad a_s := (1-\beta)\beta^{k+1-s} \quad (1 \leq s \leq k-1), \quad a_k := (1-\beta)(1+\beta).$$

96 Then the noise term equals $\sum_{s=0}^k a_s \boldsymbol{\xi}_s$. Let $\kappa_\alpha := 2\sqrt{\pi}C_\alpha^{1/\alpha}$. Using the vector-valued martingale
97 inequality, we have:

$$\mathbb{E} \left\| \sum_{s=0}^k a_s \boldsymbol{\xi}_s \right\|_F \leq 2\sqrt{\pi} \mathbb{E} \left[\left(\sum_{s=0}^k |a_s|^\alpha \|\boldsymbol{\xi}_s\|_F^\alpha \right)^{1/\alpha} \right].$$

98 Conditioning on $(\mathbf{X}_0, \dots, \mathbf{X}_k)$ and applying Jensen's inequality (since $x \mapsto x^{1/\alpha}$ is concave for
99 $\alpha > 1$), together with the generalized heavy-tailed bound:

$$\begin{aligned} \mathbb{E} \left[\left(\sum_{s=0}^k |a_s|^\alpha \|\boldsymbol{\xi}_s\|_F^\alpha \right)^{1/\alpha} \right] &\leq \mathbb{E} \left[\left(\sum_{s=0}^k |a_s|^\alpha \mathbb{E}[\|\boldsymbol{\xi}_s\|_F^\alpha \mid \mathbf{X}_s] \right)^{1/\alpha} \right] \\ &\leq \frac{1}{B^{\frac{\alpha-1}{\alpha}}} \mathbb{E} \left[\left(\sum_{s=0}^k |a_s|^\alpha (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_s)\|_F^\alpha) \right)^{1/\alpha} \right] \\ &\leq \frac{1}{B^{\frac{\alpha-1}{\alpha}}} \left[\sigma_0 \left(\sum_{s=0}^k |a_s|^\alpha \right)^{1/\alpha} + \sigma_1 \mathbb{E} \left(\sum_{s=0}^k |a_s|^\alpha \|\nabla f(\mathbf{X}_s)\|_F^\alpha \right)^{1/\alpha} \right], \end{aligned}$$

100 where the last step uses the subadditivity $(x+y)^{1/\alpha} \leq x^{1/\alpha} + y^{1/\alpha}$.

101 For the σ_1 term, we use the fact that $(\sum w_s u_s^\alpha)^{1/\alpha} \leq \sum w_s^{1/\alpha} u_s$ for any positive terms when $\alpha > 1$.

102 Setting $w_s = |a_s|^\alpha$ and $u_s = \|\nabla f(\mathbf{X}_s)\|_F$:

$$\mathbb{E} \left(\sum_{s=0}^k |a_s|^\alpha \|\nabla f(\mathbf{X}_s)\|_F^\alpha \right)^{1/\alpha} \leq \sum_{s=0}^k |a_s| \mathbb{E} \|\nabla f(\mathbf{X}_s)\|_F.$$

103 Additionally, setting $\Gamma_\alpha := (2^\alpha + 1)^{1/\alpha}$, the deterministic sum of coefficients is bounded by:

$$\left(\sum_{s=0}^k |a_s|^\alpha \right)^{1/\alpha} \leq (1-\beta) \Gamma_\alpha (1-\beta^\alpha)^{-1/\alpha} \leq (1-\beta)^{\frac{\alpha-1}{\alpha}} \Gamma_\alpha.$$

104 Combining these elements yields the bound:

$$\mathbb{E} \left\| \sum_{s=0}^k a_s \xi_s \right\|_F \leq \frac{\kappa_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \left[\Gamma_\alpha \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} \right. \quad (10)$$

$$\left. + \sigma_1 (1-\beta) \left((1+\beta) \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F + \sum_{s=1}^{k-1} \beta^{k+1-s} \mathbb{E} \|\nabla f(\mathbf{X}_s)\|_F + \beta^k \|\nabla f(\mathbf{X}_0)\|_F \right) \right]. \quad (11)$$

105 **(c) Bound $\mathbb{E} \|\mathbf{S}_k\|_F$ and sum over k .** From equation 6 and the triangle inequality, combining the
106 drift bound and equation 10 gives for all k :

$$\begin{aligned} \mathbb{E} \|\mathbf{S}_k\|_F &\leq \beta^k \mathbb{E} \|\epsilon_0\|_F + \frac{\beta^2}{1-\beta} L\eta \sqrt{d} + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} \\ &\quad + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} (1-\beta) \left((1+\beta) \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F + \sum_{s=1}^{k-1} \beta^{k+1-s} \mathbb{E} \|\nabla f(\mathbf{X}_s)\|_F + \beta^k \|\nabla f(\mathbf{X}_0)\|_F \right). \end{aligned} \quad (12)$$

107 $\square$

Theorem A.4. For $\rho := 1 - (1 + \sqrt{d}) \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}}$ we have

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq \frac{f(\mathbf{X}_0) - f_\star}{\eta K \rho} + \frac{Ld}{2\rho} \eta \\ &\quad + \frac{1 + \sqrt{d}}{\rho} \left[\frac{\mathbb{E} \|\epsilon_0\|_F}{K(1-\beta)} + \frac{\beta^2}{1-\beta} L\eta \sqrt{d} + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \frac{\|\nabla f(\mathbf{X}_0)\|_F}{K} \right]. \end{aligned}$$

108 *Proof.* We now sum equation 12 for $k = 0, \dots, K-1$. The first three terms give:

$$\sum_{k=0}^{K-1} \beta^k \mathbb{E} \|\epsilon_0\|_F \leq \frac{\mathbb{E} \|\epsilon_0\|_F}{1-\beta},$$

$$\sum_{k=0}^{K-1} \frac{\beta^2}{1-\beta} L\eta \sqrt{d} = K \frac{\beta^2}{1-\beta} L\eta \sqrt{d},$$

$$\sum_{k=0}^{K-1} \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} = K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}}.$$

110 For the σ_1 -dependent gradient-coupled part, define $S := \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F$. By swapping the
111 order of summation for the double sum and evaluating the geometric series, the coefficients on the
112 internal gradients sum exactly to $(1-\beta^2) + \beta^2 = 1$. Thus, we have the clean upper bound:

$$\sum_{k=0}^{K-1} (1-\beta) \left((1+\beta) \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F + \sum_{s=1}^{k-1} \beta^{k+1-s} \mathbb{E} \|\nabla f(\mathbf{X}_s)\|_F \right) \leq S.$$

113 The initial gradient term is similarly bounded by $\sum_{k=0}^{K-1} (1-\beta) \beta^k \|\nabla f(\mathbf{X}_0)\|_F \leq \|\nabla f(\mathbf{X}_0)\|_F$.
114 Therefore, the total error sum is bounded by:

$$\sum_{k=0}^{K-1} \mathbb{E} \|\mathbf{S}_k\|_F \leq \frac{\mathbb{E} \|\epsilon_0\|_F}{1-\beta} + K \frac{\beta^2}{1-\beta} L\eta \sqrt{d} + K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} (S + \|\nabla f(\mathbf{X}_0)\|_F). \quad (13)$$

115 Plugging equation 13 back into the one-step descent summation equation ??:

$$\begin{aligned} \eta S &\leq (f(\mathbf{X}_0) - f_\star) + \frac{Ld}{2} K \eta^2 \\ &\quad + \eta(1 + \sqrt{d}) \left[\frac{\mathbb{E} \|\epsilon_0\|_F}{1 - \beta} + K \frac{\beta^2}{1 - \beta} L \eta \sqrt{d} + K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1 - \beta)^{\frac{\alpha-1}{\alpha}} \right] \\ &\quad + \eta(1 + \sqrt{d}) \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} (S + \|\nabla f(\mathbf{X}_0)\|_F). \end{aligned}$$

116 Moving the S -term from the RHS to the LHS yields:

$$\begin{aligned} \eta \rho S &\leq (f(\mathbf{X}_0) - f_\star) + \frac{Ld}{2} K \eta^2 \\ &\quad + \eta(1 + \sqrt{d}) \left[\frac{\mathbb{E} \|\epsilon_0\|_F}{1 - \beta} + K \frac{\beta^2}{1 - \beta} L \eta \sqrt{d} + K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1 - \beta)^{\frac{\alpha-1}{\alpha}} + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \|\nabla f(\mathbf{X}_0)\|_F \right], \end{aligned}$$

117 where we define $\rho := 1 - (1 + \sqrt{d}) \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}}$. Dividing by $\eta K \rho$ to isolate the average gradient norm
118 gives:

$$\begin{aligned} \frac{1}{K} S &\leq \frac{f(\mathbf{X}_0) - f_\star}{\eta K \rho} + \frac{Ld}{2\rho} \eta \\ &\quad + \frac{1 + \sqrt{d}}{\rho} \left[\frac{\mathbb{E} \|\epsilon_0\|_F}{K(1 - \beta)} + \frac{\beta^2}{1 - \beta} L \eta \sqrt{d} + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1 - \beta)^{\frac{\alpha-1}{\alpha}} + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \frac{\|\nabla f(\mathbf{X}_0)\|_F}{K} \right]. \end{aligned}$$

119 Finally, plugging in the horizon-dependent parameters $\beta = 1 - \frac{1}{\sqrt{K}}$ and $\eta = \frac{1}{\sqrt{d} K^{3/4}}$ establishes the
120 identities:

$$\frac{1}{\eta K} = \frac{\sqrt{d}}{K^{1/4}}, \quad \frac{1}{K(1 - \beta)} = \frac{1}{K^{1/2}}, \quad \frac{\eta \sqrt{d}}{1 - \beta} = \frac{1}{K^{1/4}}, \quad (1 - \beta)^{\frac{\alpha-1}{\alpha}} = K^{-\frac{\alpha-1}{2\alpha}}.$$

121 Substituting these directly provides the exact leading rates claimed in the Theorem. $\square$

122 A.1 Sample Complexity

123 In this subsection, we derive the sample complexity from Theorem ?? . Fix any $\varepsilon > 0$. Then choose

$$B \geq \left(2\kappa_\alpha\sigma_1(1+\sqrt{d})\right)^{\alpha/\alpha-1}$$

124 to ensure $\rho \geq \frac{1}{2}$. Moreover, choose step size $\eta = \frac{1}{\sqrt{K}}$ to get

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] \leq \mathcal{O}(1/\sqrt{K}) + \mathcal{O}(B^{-(\alpha-1)/\alpha}).$$

125 Now we want the right hand side to be $\mathcal{O}(\varepsilon)$. Thus, we choose $K = \mathcal{O}(\varepsilon^{-2})$ and $B = \mathcal{O}\left(\varepsilon^{\frac{-\alpha}{\alpha-1}}\right)$ to
 126 get each term on the right hand side to be of the order $\mathcal{O}(\varepsilon)$. Therefore, the sample complexity is
 127 given by

$$\begin{aligned} K \times B &= \mathcal{O}\left(\varepsilon^{-2-\frac{\alpha}{\alpha-1}}\right) \\ &= \mathcal{O}\left(\varepsilon^{\frac{-3\alpha+2}{\alpha-1}}\right). \end{aligned}$$

128 Note that for $\alpha = 2$, this sample complexity reduces to $\mathcal{O}(\varepsilon^{-4})$, which is optimal.

129 A.2 Parameters Independent of α

130 For $\rho := 1 - (1 + \sqrt{d})^{\frac{\kappa_\alpha\sigma_1}{B^{\frac{\alpha-1}{\alpha}}}}$ we have

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq \frac{f(\mathbf{X}_0) - f_\star}{\eta K \rho} + \frac{Ld}{2\rho} \eta \\ &\quad + \frac{1 + \sqrt{d}}{\rho} \left[\frac{\mathbb{E} \|\epsilon_0\|_F}{K(1-\beta)} + \frac{\beta^2}{1-\beta} L\eta\sqrt{d} + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} \right. \\ &\quad \left. + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \frac{\|\nabla f(\mathbf{X}_0)\|_F}{K} \right]. \end{aligned}$$

131 Finally, plugging in the horizon-dependent parameters $\beta = 1 - \frac{1}{\sqrt{K}}$ and $\eta = \frac{1}{\sqrt{d}K^{3/4}}$ establishes the
 132 identities:

$$\frac{1}{\eta K} = \frac{\sqrt{d}}{K^{1/4}}, \quad \frac{1}{K(1-\beta)} = \frac{1}{K^{1/2}}, \quad \frac{\eta\sqrt{d}}{1-\beta} = \frac{1}{K^{1/4}}, \quad (1-\beta)^{\frac{\alpha-1}{\alpha}} = K^{-\frac{\alpha-1}{2\alpha}}.$$

133 Thus, we get that the sample complexity is $\mathcal{O}(\varepsilon^{-2\alpha/\alpha-1})$.

134 A.3 Inexact Polar Decomposition Analysis

135 To analyze the behavior of Muon under practical, inexact polar decomposition methods (such as
 136 Newton-Schulz iterations), we replace the exact orthogonalization $O(\mathbf{M}_k)$ with an approximation
 137 $T(\mathbf{M}_k)$. We formalize the quality of this approximation with the following assumption.

Assumption A.5 (*Inexact Polar Decomposition*). There exist constants $\delta \in [0, 1)$ and $\rho \geq 0$ such that for all iterations k , the inexact transformation $T(\mathbf{M}_k)$ satisfies:

$$\begin{aligned}\mathbb{E} [\langle \mathbf{M}_k, T(\mathbf{M}_k) \rangle \mid \mathbf{M}_k] &\geq (1 - \delta) \|\mathbf{M}_k\|_* \\ \mathbb{E} [\|T(\mathbf{M}_k)\|_{\text{op}}^2 \mid \mathbf{M}_k] &\leq (1 + \rho)^2\end{aligned}$$

138 **Note** that exact polar decomposition corresponds to $\delta = 0$ and $\rho = 0$. Because $\|T(\mathbf{M}_k)\|_F^2 \leq$
 139 $d\|T(\mathbf{M}_k)\|_{\text{op}}^2$, this assumption implies $\mathbb{E} [\|T(\mathbf{M}_k)\|_F^2 \mid \mathbf{M}_k] \leq d(1 + \rho)^2$, and by Jensen's inequality,
 140 $\mathbb{E} [\|T(\mathbf{M}_k)\|_F \mid \mathbf{M}_k] \leq \sqrt{d}(1 + \rho)$.

141 We now modify the core lemmas to account for $T(\mathbf{M}_k)$.

Lemma A.6 (Modified Descent Lemma). Under Assumption 3.1 and Assumption A.9, let $C_{\delta, \rho} := (1 - \delta) + \sqrt{d}(1 + \rho)$. Muon satisfies

$$\eta(1 - \delta) \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F \leq (f(\mathbf{X}_0) - f_*) + \frac{Ld(1 + \rho)^2}{2} K\eta^2 + \eta C_{\delta, \rho} \sum_{k=0}^{K-1} \mathbb{E} \|\mathbf{S}_k\|_F. \quad (14)$$

142 *Proof.* Using the update rule $\mathbf{X}_{k+1} = \mathbf{X}_k - \eta T(\widetilde{\mathbf{M}}_k)$ and L -smoothness, we have

$$\begin{aligned}f(\mathbf{X}_{k+1}) &\leq f(\mathbf{X}_k) + \langle \nabla f(\mathbf{X}_k), \mathbf{X}_{k+1} - \mathbf{X}_k \rangle + \frac{L}{2} \|\mathbf{X}_{k+1} - \mathbf{X}_k\|_F^2 \\ &= f(\mathbf{X}_k) - \eta \langle \nabla f(\mathbf{X}_k), T(\widetilde{\mathbf{M}}_k) \rangle + \frac{L}{2} \eta^2 \|T(\widetilde{\mathbf{M}}_k)\|_F^2.\end{aligned}$$

143 We decompose the inner product using $\widetilde{\mathbf{M}}_k = \nabla f(\mathbf{X}_k) + \mathbf{S}_k$:

$$-\langle \nabla f(\mathbf{X}_k), T(\widetilde{\mathbf{M}}_k) \rangle = -\langle \widetilde{\mathbf{M}}_k, T(\widetilde{\mathbf{M}}_k) \rangle + \langle \mathbf{S}_k, T(\widetilde{\mathbf{M}}_k) \rangle.$$

144 Taking the expectation conditioned on $\mathcal{F}_k$ (where $\widetilde{\mathbf{M}}_k$ is fixed) and applying Assumption A.9:

$$\begin{aligned}\mathbb{E} [f(\mathbf{X}_{k+1}) \mid \mathcal{F}_k] &\leq f(\mathbf{X}_k) - \eta(1 - \delta) \|\widetilde{\mathbf{M}}_k\|_* + \eta \|\mathbf{S}_k\|_F \mathbb{E} [\|T(\widetilde{\mathbf{M}}_k)\|_F \mid \mathcal{F}_k] + \frac{Ld}{2} \eta^2 (1 + \rho)^2 \\ &\leq f(\mathbf{X}_k) - \eta(1 - \delta) \|\widetilde{\mathbf{M}}_k\|_F + \eta \sqrt{d}(1 + \rho) \|\mathbf{S}_k\|_F + \frac{Ld(1 + \rho)^2}{2} \eta^2.\end{aligned}$$

145 Using $\|\widetilde{\mathbf{M}}_k\|_F \geq \|\nabla f(\mathbf{X}_k)\|_F - \|\mathbf{S}_k\|_F$ yields

$$\begin{aligned}\mathbb{E} [f(\mathbf{X}_{k+1}) \mid \mathcal{F}_k] &\leq f(\mathbf{X}_k) - \eta(1 - \delta) \|\nabla f(\mathbf{X}_k)\|_F + \eta \left((1 - \delta) + \sqrt{d}(1 + \rho) \right) \|\mathbf{S}_k\|_F + \frac{Ld(1 + \rho)^2}{2} \eta^2 \\ &= f(\mathbf{X}_k) - \eta(1 - \delta) \|\nabla f(\mathbf{X}_k)\|_F + \eta C_{\delta, \rho} \|\mathbf{S}_k\|_F + \frac{Ld(1 + \rho)^2}{2} \eta^2.\end{aligned}$$

146 Taking total expectation and summing from $k = 0$ to $K - 1$ yields the result. $\square$

147 *Note:* The unrolling of the error vector $\mathbf{S}_k$ depends purely on the Nesterov momentum recursion
 148 and the gradient estimates, not the step direction $T(\mathbf{M}_k)$. Therefore, Lemma 3 (Equation 6) holds
 149 identically for the inexact case.
 150

Lemma A.7 (Modified Error Bound).

$$\begin{aligned} \mathbb{E}\|\mathbf{S}_k\|_F &\leq \beta^k \mathbb{E}\|\boldsymbol{\epsilon}_0\|_F + \frac{\beta^2}{1-\beta} L\eta\sqrt{d}(1+\rho) + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} \\ &\quad + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} (1-\beta) \left((1+\beta) \mathbb{E}\|\nabla f(\mathbf{X}_k)\|_F + \sum_{s=1}^{k-1} \beta^{k+1-s} \mathbb{E}\|\nabla f(\mathbf{X}_s)\|_F + \beta^k \|\nabla f(\mathbf{X}_0)\|_F \right). \end{aligned} \quad (15)$$

151

152 *Proof.* The only difference from the exact polar decomposition case arises in the bound for the step
153 displacement D_s . By L -smoothness and taking expectations:

$$\mathbb{E}[\|D_s\|_F] = \mathbb{E}[\|\nabla f(\mathbf{X}_{s-1}) - \nabla f(\mathbf{X}_s)\|_F] \leq L\mathbb{E}[\|\mathbf{X}_s - \mathbf{X}_{s-1}\|_F] = L\eta\mathbb{E}[\|T(\widetilde{\mathbf{M}}_{s-1})\|_F] \leq L\eta\sqrt{d}(1+\rho).$$

154 Hence, the cumulative deterministic drift bound becomes:

$$\beta^2 \sum_{s=1}^k \beta^{k-s} \mathbb{E}\|D_s\|_F \leq \frac{\beta^2}{1-\beta} L\eta\sqrt{d}(1+\rho).$$

155 The martingale difference noise terms involving $\boldsymbol{\xi}_s$ are unaffected by the choice of $T(\mathbf{M}_k)$. Substi-
156 tuting the updated drift bound into the original proof structure completes the lemma. $\square$

Theorem A.8. Let $C_{\delta,\rho} := (1-\delta) + \sqrt{d}(1+\rho)$. For $\mu := (1-\delta) - C_{\delta,\rho} \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}}$ we have

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}[\|\nabla f(\mathbf{X}_k)\|_F] &\leq \frac{f(\mathbf{X}_0) - f_\star}{\eta K \mu} + \frac{Ld(1+\rho)^2}{2\mu} \eta \\ &\quad + \frac{C_{\delta,\rho}}{\mu} \left[\frac{\mathbb{E}\|\boldsymbol{\epsilon}_0\|_F}{K(1-\beta)} + \frac{\beta^2}{1-\beta} L\eta\sqrt{d}(1+\rho) + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \frac{\|\nabla f(\mathbf{X}_0)\|_F}{K} \right]. \end{aligned}$$

157 *Proof.* We sum equation 24 for $k = 0, \dots, K-1$. Letting $S := \sum_{k=0}^{K-1} \mathbb{E}\|\nabla f(\mathbf{X}_k)\|_F$, the total
158 error sum is bounded by:

$$\sum_{k=0}^{K-1} \mathbb{E}\|\mathbf{S}_k\|_F \leq \frac{\mathbb{E}\|\boldsymbol{\epsilon}_0\|_F}{1-\beta} + K \frac{\beta^2}{1-\beta} L\eta\sqrt{d}(1+\rho) + K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \left(S + \|\nabla f(\mathbf{X}_0)\|_F \right). \quad (16)$$

159 Plugging equation 27 into the one-step descent summation equation 22:

$$\begin{aligned} \eta(1-\delta)S &\leq (f(\mathbf{X}_0) - f_\star) + \frac{Ld(1+\rho)^2}{2} K\eta^2 \\ &\quad + \eta C_{\delta,\rho} \left[\frac{\mathbb{E}\|\boldsymbol{\epsilon}_0\|_F}{1-\beta} + K \frac{\beta^2}{1-\beta} L\eta\sqrt{d}(1+\rho) + K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} \right] \\ &\quad + \eta C_{\delta,\rho} \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \left(S + \|\nabla f(\mathbf{X}_0)\|_F \right). \end{aligned}$$

160 Moving the S -term to the left hand side yields a factor of $\eta \left((1-\delta) - C_{\delta,\rho} \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \right) S$, which equals
161 $\eta\mu S$. Dividing through by $\eta K \mu$ completes the proof. $\square$

162 A.4 Sample Complexity (Inexact Case)

163 In this subsection, we derive the sample complexity for the inexact polar decomposition. Fix any
 164 $\varepsilon > 0$. First, we must ensure $\mu \geq \frac{1-\delta}{2}$, which requires choosing a batch size:

$$B \geq \left(\frac{2\kappa_\alpha \sigma_1 C_{\delta,\rho}}{1-\delta} \right)^{\alpha/\alpha-1}.$$

165 Assuming δ and ρ are strictly bounded strictly away from 1 and infinity respectively, this only changes
 166 the batch size requirement by a constant factor relative to the exact case.

167 Plugging in the horizon-dependent parameters $\beta = 1 - \frac{1}{\sqrt{K}}$ and $\eta = \frac{1}{\sqrt{d}K^{3/4}}$, we get

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] \leq \mathcal{O}(1/K^{1/4}) + \mathcal{O}(B^{-(\alpha-1)/\alpha}).$$

168 (Note: Using $\eta = \frac{1}{\sqrt{d}K^{3/4}}$ leads to the $\mathcal{O}(K^{-1/4})$ rate as established in the original identity
 169 substitutions. If $\eta = \frac{1}{\sqrt{K}}$ is used instead as mentioned in your original Sample Complexity section,
 170 the rate is $\mathcal{O}(1/\sqrt{K})$.)

171 To bound the right hand side by $\mathcal{O}(\varepsilon)$, we choose $K = \mathcal{O}(\varepsilon^{-4})$ and $B = \mathcal{O}\left(\varepsilon^{\frac{-\alpha}{\alpha-1}}\right)$. Therefore, the
 172 sample complexity scales as:

$$\begin{aligned} K \times B &= \mathcal{O}\left(\varepsilon^{-4 - \frac{\alpha}{\alpha-1}}\right) \\ &= \mathcal{O}\left(\varepsilon^{\frac{-5\alpha+4}{\alpha-1}}\right). \end{aligned}$$

173 Crucially, the asymptotic dependencies on ε are identical to the exact polar decomposition case; the
 174 inexactness penalties δ and ρ merely scale the required constants.

175 A.5 Muon with Nesterov momentum and inexact polar decomposition

176 In this subsection, we analyze the constant-parameter specialization

$$\mathbf{G}_k = \frac{1}{B} \sum_{i=1}^B \mathbf{G}_k^i, \quad (17)$$

$$\mathbf{C}_k = \beta \mathbf{C}_{k-1} + \mathbf{G}_k, \quad (18)$$

$$\mathbf{M}_k = \beta \mathbf{C}_k + \mathbf{G}_k, \quad (19)$$

$$\mathbf{X}_{k+1} = \mathbf{X}_k - \eta T(\mathbf{M}_k), \quad (20)$$

177 where $\beta \in (0, 1)$ and $\eta > 0$ are constants.

178 We use the same heavy-tailed mini-batch assumption as in the exact-polar case, and we replace the
179 exact polar map by the following inexact oracle assumption.

Assumption A.9 (Inexact polar decomposition). For each k , the (possibly randomized) map $T(\mathbf{M}_k)$ satisfies

$$\begin{aligned} \mathbb{E}[\langle \mathbf{M}_k, T(\mathbf{M}_k) \rangle \mid \mathbf{M}_k] &\geq (1 - \delta_k) \|\mathbf{M}_k\|_*, \\ \mathbb{E}[\|T(\mathbf{M}_k)\|^2 \mid \mathbf{M}_k] &\leq (1 + \rho_k)^2, \end{aligned}$$

where $\delta_k \in [0, 1)$ and $\rho_k \geq 0$.

180 As before, define the rescaled momentum variables

$$\tilde{\mathbf{C}}_k := (1 - \beta) \mathbf{C}_k, \quad \tilde{\mathbf{M}}_k := (1 - \beta) \mathbf{M}_k,$$

181 and the stochastic-gradient and momentum errors

$$\boldsymbol{\xi}_k := \mathbf{G}_k - \nabla f(\mathbf{X}_k), \quad \boldsymbol{\epsilon}_k := \tilde{\mathbf{M}}_k - \nabla f(\mathbf{X}_k).$$

182 We first record the mini-batch noise bound, which is identical to the exact-polar case.

Lemma A.10. Under Assumption 3.2, we have

$$\mathbb{E}[\|\boldsymbol{\xi}_k\|_F^\alpha \mid \mathbf{X}_k] \leq \frac{C_\alpha}{B^{\alpha-1}} (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_k)\|_F^\alpha) \quad (21)$$

for some constant $C_\alpha > 0$ independent of B .

183 *Proof.* By definition,

$$\boldsymbol{\xi}_k = \mathbf{G}_k - \nabla f(\mathbf{X}_k) = \frac{1}{B} \sum_{i=1}^B (\mathbf{G}_k^i - \nabla f(\mathbf{X}_k)).$$

184 Hence

$$\begin{aligned} \mathbb{E}[\|\boldsymbol{\xi}_k\|_F^\alpha \mid \mathbf{X}_k] &= \frac{1}{B^\alpha} \mathbb{E} \left[\left\| \sum_{i=1}^B (\mathbf{G}_k^i - \nabla f(\mathbf{X}_k)) \right\|_F^\alpha \mid \mathbf{X}_k \right] \\ &\leq \frac{C_\alpha}{B^\alpha} \sum_{i=1}^B \mathbb{E}[\|\mathbf{G}_k^i - \nabla f(\mathbf{X}_k)\|_F^\alpha \mid \mathbf{X}_k], \end{aligned}$$

185 where the inequality follows from the Bahr–Esseen inequality applied conditionally on $\mathbf{X}_k$. Using
186 Assumption 3.2 gives

$$\begin{aligned} \mathbb{E}[\|\boldsymbol{\xi}_k\|_F^\alpha \mid \mathbf{X}_k] &\leq \frac{C_\alpha}{B^\alpha} \sum_{i=1}^B (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_k)\|_F^\alpha) \\ &= \frac{C_\alpha}{B^{\alpha-1}} (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_k)\|_F^\alpha). \end{aligned}$$

187 □

188 The main new ingredient is the basic descent inequality under the inexact polar oracle.

Lemma A.11. Let

$$\bar{\delta} := \max_{0 \leq k \leq K-1} \delta_k, \quad \bar{\rho} := \max_{0 \leq k \leq K-1} \rho_k.$$

Under Assumptions 3.1 and A.9,

$$\begin{aligned} \eta(1 - \bar{\delta}) \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F &\leq (f(\mathbf{X}_0) - f_\star) + \frac{Ld}{2} K \eta^2 (1 + \bar{\rho})^2 \\ &\quad + \eta(1 + \sqrt{d}(1 + \bar{\rho})) \sum_{k=0}^{K-1} \mathbb{E} \|\epsilon_k\|_F. \end{aligned} \quad (22)$$

189 *Proof.* Let $\mathcal{G}_k$ be the sigma-field generated by the algorithm history up to the moment when $\mathbf{M}_k$ has
 190 been formed, but before the randomized oracle $T(\mathbf{M}_k)$ is sampled. Then $\mathbf{X}_k$, $\mathbf{M}_k$, $\widetilde{\mathbf{M}}_k$, and ϵ_k are
 191 $\mathcal{G}_k$ -measurable.

192 By L -smoothness,

$$\begin{aligned} f(\mathbf{X}_{k+1}) &\leq f(\mathbf{X}_k) + \langle \nabla f(\mathbf{X}_k), \mathbf{X}_{k+1} - \mathbf{X}_k \rangle + \frac{L}{2} \|\mathbf{X}_{k+1} - \mathbf{X}_k\|_F^2 \\ &= f(\mathbf{X}_k) - \eta \langle \nabla f(\mathbf{X}_k), T(\mathbf{M}_k) \rangle + \frac{L\eta^2}{2} \|T(\mathbf{M}_k)\|_F^2. \end{aligned}$$

193 Since $\widetilde{\mathbf{M}}_k = \nabla f(\mathbf{X}_k) + \epsilon_k$, we have

$$f(\mathbf{X}_{k+1}) \leq f(\mathbf{X}_k) - \eta \langle \widetilde{\mathbf{M}}_k, T(\mathbf{M}_k) \rangle + \eta \langle \epsilon_k, T(\mathbf{M}_k) \rangle + \frac{L\eta^2}{2} \|T(\mathbf{M}_k)\|_F^2.$$

194 Taking conditional expectation given $\mathcal{G}_k$, and using $\widetilde{\mathbf{M}}_k = (1 - \beta)\mathbf{M}_k$, yields

$$\begin{aligned} \mathbb{E}[f(\mathbf{X}_{k+1}) \mid \mathcal{G}_k] &\leq f(\mathbf{X}_k) - \eta \mathbb{E}[\langle \widetilde{\mathbf{M}}_k, T(\mathbf{M}_k) \rangle \mid \mathcal{G}_k] \\ &\quad + \eta \mathbb{E}[\langle \epsilon_k, T(\mathbf{M}_k) \rangle \mid \mathcal{G}_k] + \frac{L\eta^2}{2} \mathbb{E}[\|T(\mathbf{M}_k)\|_F^2 \mid \mathcal{G}_k]. \end{aligned}$$

195 For the first term,

$$\begin{aligned} \mathbb{E}[\langle \widetilde{\mathbf{M}}_k, T(\mathbf{M}_k) \rangle \mid \mathcal{G}_k] &= (1 - \beta) \mathbb{E}[\langle \mathbf{M}_k, T(\mathbf{M}_k) \rangle \mid \mathcal{G}_k] \\ &= (1 - \beta) \mathbb{E}[\mathbb{E}[\langle \mathbf{M}_k, T(\mathbf{M}_k) \rangle \mid \mathbf{M}_k] \mid \mathcal{G}_k] \\ &\geq (1 - \beta)(1 - \delta_k) \|\mathbf{M}_k\|_* \\ &= (1 - \delta_k) \|\widetilde{\mathbf{M}}_k\|_*. \end{aligned}$$

196 For the second term, by Cauchy–Schwarz and Jensen,

$$\begin{aligned} \mathbb{E}[\langle \epsilon_k, T(\mathbf{M}_k) \rangle \mid \mathcal{G}_k] &\leq \|\epsilon_k\|_F \mathbb{E}[\|T(\mathbf{M}_k)\|_F \mid \mathcal{G}_k] \\ &\leq \|\epsilon_k\|_F \left(\mathbb{E}[\|T(\mathbf{M}_k)\|_F^2 \mid \mathcal{G}_k] \right)^{1/2}. \end{aligned}$$

197 Using $\|A\|_F^2 \leq d\|A\|^2$ for $d \times d$ matrices and Assumption A.9,

$$\mathbb{E}[\|T(\mathbf{M}_k)\|_F^2 \mid \mathcal{G}_k] \leq d \mathbb{E}[\|T(\mathbf{M}_k)\|^2 \mid \mathcal{G}_k] \leq d(1 + \rho_k)^2.$$

198 Hence

$$\mathbb{E}[\langle \epsilon_k, T(\mathbf{M}_k) \rangle \mid \mathcal{G}_k] \leq \sqrt{d}(1 + \rho_k) \|\epsilon_k\|_F.$$

199 Combining the above bounds gives

$$\begin{aligned} \mathbb{E}[f(\mathbf{X}_{k+1}) \mid \mathcal{G}_k] &\leq f(\mathbf{X}_k) - \eta(1 - \delta_k) \|\widetilde{\mathbf{M}}_k\|_* + \eta\sqrt{d}(1 + \rho_k) \|\epsilon_k\|_F \\ &\quad + \frac{Ld}{2} \eta^2 (1 + \rho_k)^2. \end{aligned}$$

200 Since $\|\widetilde{\mathbf{M}}_k\|_* \geq \|\widetilde{\mathbf{M}}_k\|_F$ and

$$\|\widetilde{\mathbf{M}}_k\|_F = \|\nabla f(\mathbf{X}_k) + \boldsymbol{\epsilon}_k\|_F \geq \|\nabla f(\mathbf{X}_k)\|_F - \|\boldsymbol{\epsilon}_k\|_F,$$

201 we obtain

$$\begin{aligned} \mathbb{E}[f(\mathbf{X}_{k+1}) \mid \mathcal{G}_k] &\leq f(\mathbf{X}_k) - \eta(1 - \delta_k)\|\nabla f(\mathbf{X}_k)\|_F \\ &\quad + \eta((1 - \delta_k) + \sqrt{d}(1 + \rho_k))\|\boldsymbol{\epsilon}_k\|_F + \frac{Ld}{2}\eta^2(1 + \rho_k)^2. \end{aligned}$$

202 Taking total expectation and summing over $k = 0, \dots, K - 1$, then using

$$1 - \delta_k \geq 1 - \bar{\delta}, \quad (1 - \delta_k) + \sqrt{d}(1 + \rho_k) \leq 1 + \sqrt{d}(1 + \bar{\rho}), \quad (1 + \rho_k)^2 \leq (1 + \bar{\rho})^2,$$

203 yields equation 22. $\square$

204 Next, the recursion for $\boldsymbol{\epsilon}_k$ is unchanged, because it depends only on the momentum update and not
205 on whether the polar oracle is exact or inexact.

Lemma A.12. For all $k \geq 1$,

$$\boldsymbol{\epsilon}_k = \beta^k \boldsymbol{\epsilon}_0 + \beta^2 \sum_{s=1}^k \beta^{k-s} D_s + (1 - \beta) \left((1 + \beta) \boldsymbol{\xi}_k + \sum_{s=1}^{k-1} \beta^{k+1-s} \boldsymbol{\xi}_s - \beta^k \boldsymbol{\xi}_0 \right), \quad (23)$$

where

$$D_s := \nabla f(\mathbf{X}_{s-1}) - \nabla f(\mathbf{X}_s).$$

206 *Proof.* Using

$$\widetilde{\mathbf{C}}_k = \beta \widetilde{\mathbf{C}}_{k-1} + (1 - \beta) \mathbf{G}_k, \quad \widetilde{\mathbf{M}}_k = \beta \widetilde{\mathbf{C}}_k + (1 - \beta) \mathbf{G}_k,$$

207 we obtain

$$\begin{aligned} \widetilde{\mathbf{M}}_k &= \beta \widetilde{\mathbf{C}}_k + (1 - \beta) \mathbf{G}_k \\ &= \beta(\beta \widetilde{\mathbf{C}}_{k-1} + (1 - \beta) \mathbf{G}_k) + (1 - \beta) \mathbf{G}_k \\ &= \beta^2 \widetilde{\mathbf{C}}_{k-1} + (1 - \beta)(1 + \beta) \mathbf{G}_k. \end{aligned}$$

208 Also,

$$\widetilde{\mathbf{M}}_{k-1} = \beta \widetilde{\mathbf{C}}_{k-1} + (1 - \beta) \mathbf{G}_{k-1},$$

209 hence

$$\beta \widetilde{\mathbf{C}}_{k-1} = \widetilde{\mathbf{M}}_{k-1} - (1 - \beta) \mathbf{G}_{k-1}.$$

210 Substituting this gives

$$\widetilde{\mathbf{M}}_k = \beta \widetilde{\mathbf{M}}_{k-1} + (1 - \beta)((1 + \beta) \mathbf{G}_k - \beta \mathbf{G}_{k-1}).$$

211 Subtracting $\nabla f(\mathbf{X}_k)$ and using $\mathbf{G}_j = \nabla f(\mathbf{X}_j) + \boldsymbol{\xi}_j$, we get

$$\begin{aligned} \boldsymbol{\epsilon}_k &= \widetilde{\mathbf{M}}_k - \nabla f(\mathbf{X}_k) \\ &= \beta(\boldsymbol{\epsilon}_{k-1} + \nabla f(\mathbf{X}_{k-1})) + (1 - \beta)((1 + \beta)(\nabla f(\mathbf{X}_k) + \boldsymbol{\xi}_k) - \beta(\nabla f(\mathbf{X}_{k-1}) + \boldsymbol{\xi}_{k-1})) - \nabla f(\mathbf{X}_k) \\ &= \beta \boldsymbol{\epsilon}_{k-1} + \beta^2(\nabla f(\mathbf{X}_{k-1}) - \nabla f(\mathbf{X}_k)) + (1 - \beta)((1 + \beta) \boldsymbol{\xi}_k - \beta \boldsymbol{\xi}_{k-1}) \\ &= \beta \boldsymbol{\epsilon}_{k-1} + \beta^2 D_k + (1 - \beta)((1 + \beta) \boldsymbol{\xi}_k - \beta \boldsymbol{\xi}_{k-1}). \end{aligned}$$

212 Unrolling this recursion yields equation 23. $\square$

213 We now bound $\mathbb{E}\|\boldsymbol{\epsilon}_k\|_F$. The stochastic part is identical to the exact-polar case; the only change is
214 the bound on the drift term D_s , because the step length is no longer deterministic.

Lemma A.13. Let

$$\kappa_\alpha := 2\sqrt{\pi} C_\alpha^{1/\alpha}, \quad \Gamma_\alpha := (2^\alpha + 1)^{1/\alpha}.$$

Under Assumptions 3.1, 3.2, and A.9, for every $k \geq 0$,

$$\begin{aligned} \mathbb{E}\|\epsilon_k\|_F &\leq \beta^k \mathbb{E}\|\epsilon_0\|_F + \frac{\beta^2}{1-\beta} L\eta\sqrt{d}(1+\bar{\rho}) + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} \\ &\quad + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} (1-\beta) \left((1+\beta) \mathbb{E}\|\nabla f(\mathbf{X}_k)\|_F + \sum_{s=1}^{k-1} \beta^{k+1-s} \mathbb{E}\|\nabla f(\mathbf{X}_s)\|_F + \beta^k \|\nabla f(\mathbf{X}_0)\|_F \right). \end{aligned} \quad (24)$$

215 *Proof.* From equation 23,

$$\epsilon_k = \beta^k \epsilon_0 + \beta^2 \sum_{s=1}^k \beta^{k-s} D_s + (1-\beta) \left((1+\beta) \xi_k + \sum_{s=1}^{k-1} \beta^{k+1-s} \xi_s - \beta^k \xi_0 \right).$$

216 We bound the deterministic drift term first. By L -smoothness,

$$\|D_s\|_F = \|\nabla f(\mathbf{X}_{s-1}) - \nabla f(\mathbf{X}_s)\|_F \leq L \|\mathbf{X}_s - \mathbf{X}_{s-1}\|_F = L\eta \|T(\mathbf{M}_{s-1})\|_F.$$

217 Taking expectation and using Cauchy–Schwarz, $\|A\|_F \leq \sqrt{d} \|A\|$, and Assumption A.9,

$$\begin{aligned} \mathbb{E}\|D_s\|_F &\leq L\eta \mathbb{E}\|T(\mathbf{M}_{s-1})\|_F \\ &\leq L\eta \left(\mathbb{E}\|T(\mathbf{M}_{s-1})\|_F^2 \right)^{1/2} \\ &\leq L\eta\sqrt{d} \left(\mathbb{E}\|T(\mathbf{M}_{s-1})\|^2 \right)^{1/2} \\ &\leq L\eta\sqrt{d} (1 + \rho_{s-1}) \leq L\eta\sqrt{d} (1 + \bar{\rho}). \end{aligned}$$

218 Therefore,

$$\begin{aligned} \beta^2 \sum_{s=1}^k \beta^{k-s} \mathbb{E}\|D_s\|_F &\leq \beta^2 L\eta\sqrt{d} (1 + \bar{\rho}) \sum_{j=0}^{k-1} \beta^j \\ &\leq \frac{\beta^2}{1-\beta} L\eta\sqrt{d} (1 + \bar{\rho}). \end{aligned}$$

219 Now define

$$a_0 := -(1-\beta)\beta^k, \quad a_s := (1-\beta)\beta^{k+1-s} \quad (1 \leq s \leq k-1), \quad a_k := (1-\beta)(1+\beta).$$

220 Then the stochastic term in equation 23 is $\sum_{s=0}^k a_s \xi_s$. Since $\{\xi_s\}$ is a martingale difference sequence,
221 the same vector-valued martingale inequality as in the exact-polar proof gives

$$\mathbb{E} \left\| \sum_{s=0}^k a_s \xi_s \right\|_F \leq 2\sqrt{\pi} \mathbb{E} \left[\left(\sum_{s=0}^k |a_s|^\alpha \|\xi_s\|_F^\alpha \right)^{1/\alpha} \right].$$

222 Conditioning on $(\mathbf{X}_0, \dots, \mathbf{X}_k)$, using Jensen’s inequality and equation 21, we obtain

$$\begin{aligned} \mathbb{E} \left\| \sum_{s=0}^k a_s \xi_s \right\|_F &\leq \frac{\kappa_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \mathbb{E} \left[\left(\sum_{s=0}^k |a_s|^\alpha (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_s)\|_F^\alpha) \right)^{1/\alpha} \right] \\ &\leq \frac{\kappa_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \left[\sigma_0 \left(\sum_{s=0}^k |a_s|^\alpha \right)^{1/\alpha} + \sigma_1 \mathbb{E} \left(\sum_{s=0}^k |a_s|^\alpha \|\nabla f(\mathbf{X}_s)\|_F^\alpha \right)^{1/\alpha} \right]. \end{aligned}$$

223 Using $(\sum w_s u_s^\alpha)^{1/\alpha} \leq \sum w_s^{1/\alpha} u_s$ for $w_s, u_s \geq 0$, we get

$$\mathbb{E} \left(\sum_{s=0}^k |a_s|^\alpha \|\nabla f(\mathbf{X}_s)\|_F^\alpha \right)^{1/\alpha} \leq \sum_{s=0}^k |a_s| \mathbb{E} \|\nabla f(\mathbf{X}_s)\|_F.$$

224 Also,

$$\begin{aligned}
\left(\sum_{s=0}^k |a_s|^\alpha\right)^{1/\alpha} &\leq (1-\beta) \left((1+\beta)^\alpha + \sum_{j=1}^k \beta^{\alpha j}\right)^{1/\alpha} \\
&\leq (1-\beta) \left(2^\alpha + \frac{1}{1-\beta^\alpha}\right)^{1/\alpha} \\
&\leq (1-\beta) \Gamma_\alpha (1-\beta^\alpha)^{-1/\alpha} \\
&\leq \Gamma_\alpha (1-\beta)^{\frac{\alpha-1}{\alpha}}.
\end{aligned}$$

225 Therefore,

$$\begin{aligned}
\mathbb{E} \left\| \sum_{s=0}^k a_s \xi_s \right\|_F &\leq \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} \\
&\quad + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} (1-\beta) \left((1+\beta) \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F + \sum_{s=1}^{k-1} \beta^{k+1-s} \mathbb{E} \|\nabla f(\mathbf{X}_s)\|_F + \beta^k \|\nabla f(\mathbf{X}_0)\|_F \right).
\end{aligned} \tag{25}$$

226 Finally, applying the triangle inequality to equation 23, and combining the drift bound with equation 25, yields equation 24. $\square$

228 We now sum the error bound and combine it with the basic descent inequality.

Theorem A.14. Let

$$\bar{\delta} := \max_{0 \leq k \leq K-1} \delta_k, \quad \bar{\rho} := \max_{0 \leq k \leq K-1} \rho_k, \quad A_\rho := 1 + \sqrt{d}(1 + \bar{\rho}),$$

and define

$$\Theta_{\delta, \rho} := (1 - \bar{\delta}) - A_\rho \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}}.$$

If $\Theta_{\delta, \rho} > 0$, then

$$\begin{aligned}
\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F &\leq \frac{f(\mathbf{X}_0) - f_\star}{\eta K \Theta_{\delta, \rho}} + \frac{Ld}{2\Theta_{\delta, \rho}} (1 + \bar{\rho})^2 \eta \\
&\quad + \frac{A_\rho}{\Theta_{\delta, \rho}} \left[\frac{\mathbb{E} \|\epsilon_0\|_F}{K(1-\beta)} + \frac{\beta^2}{1-\beta} L\eta \sqrt{d} (1 + \bar{\rho}) + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \frac{\|\nabla f(\mathbf{X}_0)\|_F}{K} \right].
\end{aligned} \tag{26}$$

229 *Proof.* Let

$$S := \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F.$$

230 Summing equation 24 over $k = 0, \dots, K-1$, we obtain

$$\begin{aligned}
\sum_{k=0}^{K-1} \mathbb{E} \|\epsilon_k\|_F &\leq \sum_{k=0}^{K-1} \beta^k \mathbb{E} \|\epsilon_0\|_F + K \frac{\beta^2}{1-\beta} L\eta \sqrt{d} (1 + \bar{\rho}) + K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} \\
&\quad + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \sum_{k=0}^{K-1} (1-\beta) \left((1+\beta) \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F + \sum_{s=1}^{k-1} \beta^{k+1-s} \mathbb{E} \|\nabla f(\mathbf{X}_s)\|_F + \beta^k \|\nabla f(\mathbf{X}_0)\|_F \right).
\end{aligned}$$

231 The first term is bounded by

$$\sum_{k=0}^{K-1} \beta^k \mathbb{E} \|\epsilon_0\|_F \leq \frac{\mathbb{E} \|\epsilon_0\|_F}{1-\beta}.$$

232 For the gradient-coupled term, observe that for each $s \in \{1, \dots, K-1\}$, the total coefficient of
 233 $\mathbb{E}\|\nabla f(\mathbf{X}_s)\|_F$ is

$$(1-\beta)(1+\beta) + (1-\beta) \sum_{j=2}^{K-s} \beta^j \leq (1-\beta)(1+\beta) + \beta^2 = 1.$$

234 Hence

$$\sum_{k=0}^{K-1} (1-\beta) \left((1+\beta) \mathbb{E}\|\nabla f(\mathbf{X}_k)\|_F + \sum_{s=1}^{k-1} \beta^{k+1-s} \mathbb{E}\|\nabla f(\mathbf{X}_s)\|_F \right) \leq S.$$

235 Also,

$$\sum_{k=0}^{K-1} (1-\beta) \beta^k \|\nabla f(\mathbf{X}_0)\|_F \leq \|\nabla f(\mathbf{X}_0)\|_F.$$

236 Therefore,

$$\sum_{k=0}^{K-1} \mathbb{E}\|\epsilon_k\|_F \leq \frac{\mathbb{E}\|\epsilon_0\|_F}{1-\beta} + K \frac{\beta^2}{1-\beta} L \eta \sqrt{d} (1+\bar{\rho}) + K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \left(S + \|\nabla f(\mathbf{X}_0)\|_F \right). \quad (27)$$

237 Now substitute equation 27 into the basic descent bound equation 22:

$$\begin{aligned} \eta(1-\bar{\delta})S &\leq (f(\mathbf{X}_0) - f_\star) + \frac{Ld}{2} K \eta^2 (1+\bar{\rho})^2 \\ &\quad + \eta A_\rho \left[\frac{\mathbb{E}\|\epsilon_0\|_F}{1-\beta} + K \frac{\beta^2}{1-\beta} L \eta \sqrt{d} (1+\bar{\rho}) + K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} \right] \\ &\quad + \eta A_\rho \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \left(S + \|\nabla f(\mathbf{X}_0)\|_F \right). \end{aligned}$$

238 Move the S -term on the right-hand side to the left:

$$\begin{aligned} \eta \Theta_{\delta, \rho} S &\leq (f(\mathbf{X}_0) - f_\star) + \frac{Ld}{2} K \eta^2 (1+\bar{\rho})^2 \\ &\quad + \eta A_\rho \left[\frac{\mathbb{E}\|\epsilon_0\|_F}{1-\beta} + K \frac{\beta^2}{1-\beta} L \eta \sqrt{d} (1+\bar{\rho}) + K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \|\nabla f(\mathbf{X}_0)\|_F \right]. \end{aligned}$$

239 Finally, divide by $\eta K \Theta_{\delta, \rho}$ to obtain equation 26. $\square$

Remark A.15. Setting $\delta_k = 0$ and $\rho_k = 0$ for all k gives

$$A_\rho = 1 + \sqrt{d}, \quad \Theta_{\delta, \rho} = 1 - (1 + \sqrt{d}) \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}},$$

so Theorem A.14 reduces exactly to the exact-polar bound.

240 A.6 Sample complexity under uniformly bounded inexactness

241 The theorem above implies the same sample-complexity order as in the exact-polar case, up to
 242 constant factors depending on the inexactness parameters.

243 **Corollary A.16.** Assume that

$$\delta_k \leq \bar{\delta} < 1, \quad \rho_k \leq \bar{\rho} \quad \text{for all } k \geq 0,$$

and let

$$A_\rho = 1 + \sqrt{d}(1 + \bar{\rho}).$$

If the batch size satisfies

$$B \geq \left(\frac{2A_\rho \kappa_\alpha \sigma_1}{1 - \bar{\delta}} \right)^{\frac{\alpha}{\alpha-1}},$$

then

$$\Theta_{\delta, \rho} \geq \frac{1 - \bar{\delta}}{2}.$$

If $\beta \in (0, 1)$ is fixed independently of K and $\eta = K^{-1/2}$, then

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F = \mathcal{O}(K^{-1/2}) + \mathcal{O}\left(B^{-\frac{\alpha-1}{\alpha}}\right),$$

where the hidden constants depend on

$$L, d, \beta, \bar{\delta}, \bar{\rho}, \sigma_0, \sigma_1, \mathbb{E} \|\epsilon_0\|_F, \|\nabla f(\mathbf{X}_0)\|_F, f(\mathbf{X}_0) - f_\star.$$

Consequently, choosing

$$K = \mathcal{O}(\varepsilon^{-2}), \quad B = \mathcal{O}(\varepsilon^{-\frac{\alpha}{\alpha-1}})$$

yields

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F = \mathcal{O}(\varepsilon),$$

and the sample complexity is

$$KB = \mathcal{O}(\varepsilon^{-2 - \frac{\alpha}{\alpha-1}}) = \mathcal{O}(\varepsilon^{-\frac{3\alpha-2}{\alpha-1}}).$$

244

245 *Proof.* The lower bound on B implies

$$A_\rho \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \leq \frac{1 - \bar{\delta}}{2},$$

246 hence $\Theta_{\delta, \rho} \geq (1 - \bar{\delta})/2$.

247 Now substitute $\eta = K^{-1/2}$ into equation 26. Since β is fixed,

$$\frac{1}{\eta K} = K^{-1/2}, \quad \frac{1}{K(1 - \beta)} = \mathcal{O}(K^{-1}), \quad \frac{\eta}{1 - \beta} = \mathcal{O}(K^{-1/2}), \quad (1 - \beta)^{\frac{\alpha-1}{\alpha}} = \Theta(1).$$

248 Hence every K -dependent term is at most order $K^{-1/2}$, while the only batch-dependent floor is order

249 $B^{-(\alpha-1)/\alpha}$. Thus

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F = \mathcal{O}(K^{-1/2}) + \mathcal{O}\left(B^{-\frac{\alpha-1}{\alpha}}\right).$$

250 To make the right-hand side $\mathcal{O}(\varepsilon)$, choose

$$K = \mathcal{O}(\varepsilon^{-2}), \quad B = \mathcal{O}(\varepsilon^{-\frac{\alpha}{\alpha-1}}).$$

251 Multiplying gives the claimed sample complexity. $\square$

252 A.7 Horizon-dependent parameters

253 A direct analogue of the horizon-dependent choice in the exact-polar analysis is the following.

Corollary A.17. Assume $\Theta_{\delta,\rho} > 0$ is bounded below by a positive constant independent of K . Choose

$$\beta = 1 - \frac{1}{\sqrt{K}}, \quad \eta = \frac{1}{(1 + \bar{\rho})\sqrt{d} K^{3/4}}.$$

Then

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F = \mathcal{O}(K^{-1/4}) + \mathcal{O}\left(\frac{1}{B^{\frac{\alpha-1}{\alpha}} K^{\frac{\alpha-1}{2\alpha}}}\right).$$

In particular, for $B = 1$,

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F = \mathcal{O}\left(K^{-\frac{\alpha-1}{2\alpha}}\right),$$

and hence

$$K = \mathcal{O}\left(\varepsilon^{-\frac{2\alpha}{\alpha-1}}\right).$$

For $\alpha = 2$, this becomes $K = \mathcal{O}(\varepsilon^{-4})$.

254 *Proof.* Under the stated choice of β and η ,

$$\frac{1}{\eta K} = (1 + \bar{\rho})\sqrt{d} K^{-1/4}, \quad \frac{1}{K(1 - \beta)} = K^{-1/2}, \quad \frac{\eta\sqrt{d}(1 + \bar{\rho})}{1 - \beta} = K^{-1/4}, \quad (1 - \beta)^{\frac{\alpha-1}{\alpha}} = K^{-\frac{\alpha-1}{2\alpha}}.$$

255 Substituting these identities into equation 26 shows that the dominant terms are

$$\mathcal{O}(K^{-1/4}) \quad \text{and} \quad \mathcal{O}\left(\frac{1}{B^{\frac{\alpha-1}{\alpha}} K^{\frac{\alpha-1}{2\alpha}}}\right),$$

256 while the remaining terms are of smaller order ($K^{-1/2}$, $K^{-3/4}$, or $B^{-(\alpha-1)/\alpha} K^{-1}$). This proves
 257 the claim. $\square$

258 B Inexact Polar Decomposition

259 Consider the singular value decomposition of $\mathbf{M} \in \mathbb{R}^{m \times n}$

$$\begin{aligned}\mathbf{M} &= \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top, \\ \mathbf{\Sigma} &= \text{diag}(\sigma_1, \dots, \sigma_r)\end{aligned}$$

260 with $\sigma_1 \geq \dots \geq \sigma_r > 0$, where $r = \text{rank}(\mathbf{M})$. Then, the polar decomposition of $\mathbf{M}$ is given by

$$\mathbf{O}(\mathbf{M}) := \mathbf{U} \mathbf{V}^\top.$$

261 We say that an approximation $T(\mathbf{M})$ satisfies the inexact polar assumptions if there exist $\delta(\mathbf{M}) \in$
262 $[0, 1]$ and $\rho(\mathbf{M}) \geq 0$ such that

$$\begin{aligned}\langle \mathbf{M}, T(\mathbf{M}) \rangle &\geq (1 - \delta(\mathbf{M})) \|\mathbf{M}\|_*, \\ \|T(\mathbf{M})\|_{\text{op}} &\leq 1 + \rho(\mathbf{M}).\end{aligned}$$

263 Here, we show that the approximation we obtain through a few iterations of Newton Schulz satisfies
264 this assumption.

265 **Newton–Schulz approximation.** Fix a scalar $\alpha \geq \|\mathbf{M}\|_{\text{op}}$ and define the normalized cubic
266 Newton–Schulz iterates by

$$\mathbf{X}_{t+1} = \frac{3}{2} \mathbf{X}_t - \frac{1}{2} \mathbf{X}_t \mathbf{X}_t^\top \mathbf{X}_t, \quad t = 0, 1, \dots, q-1.$$

267 where $\mathbf{X}_0 = \frac{\mathbf{M}}{\alpha}$. Now, set $T_q(\mathbf{M}) := \mathbf{X}_q$ and we will show $T_q(\mathbf{M})$ satisfies the assumptions.

Theorem B.1. Define $s_i := \frac{\sigma_i}{\alpha} \in (0, 1], i = 1, \dots, r$ and let

$$f(x) := \frac{3}{2}x - \frac{1}{2}x^3.$$

Moreover, define the sequence of polynomials $p_0(x) := x$ and $p_{t+1}(x) := f(p_t(x))$. Then the Newton–Schulz iterate $T_q(\mathbf{M})$ admits the representation

$$T_q(\mathbf{M}) = \mathbf{U} \text{diag}(p_q(s_1), \dots, p_q(s_r)) \mathbf{V}^\top.$$

Consequently, $T_q(\mathbf{M})$ satisfies the assumptions with the parameters

$$\begin{aligned}\rho_q(\mathbf{M}) &= 0, \\ \delta_q(\mathbf{M}) &= 1 - \frac{\sum_{i=1}^r \sigma_i p_q(s_i)}{\sum_{i=1}^r \sigma_i}.\end{aligned}$$

Equivalently,

$$\begin{aligned}\langle \mathbf{M}, T_q(\mathbf{M}) \rangle &= (1 - \delta_q(\mathbf{M})) \|\mathbf{M}\|_*, \\ \|T_q(\mathbf{M})\|_{\text{op}} &\leq 1.\end{aligned}$$

268 *Proof.* First, we show for every $t \geq 0$,

$$\mathbf{X}_t = \mathbf{U} \text{diag}(p_t(s_1), \dots, p_t(s_r)) \mathbf{V}^\top.$$

269 We prove this by induction on t . For $t = 0$, this is immediate from

$$\begin{aligned}\mathbf{X}_0 = \frac{\mathbf{M}}{\alpha} &= \mathbf{U} \text{diag}\left(\frac{\sigma_1}{\alpha}, \dots, \frac{\sigma_r}{\alpha}\right) \mathbf{V}^\top \\ &= \mathbf{U} \text{diag}(s_1, \dots, s_r) \mathbf{V}^\top.\end{aligned}$$

270 Assume the representation holds at time t , i.e. $\mathbf{X}_t = \mathbf{U} \mathbf{D}_t \mathbf{V}^\top$, where

$$\mathbf{D}_t := \text{diag}(p_t(s_1), \dots, p_t(s_r)).$$

271 Then $\mathbf{X}_t \mathbf{X}_t^\top \mathbf{X}_t = (\mathbf{U} \mathbf{D}_t \mathbf{V}^\top)(\mathbf{V} \mathbf{D}_t \mathbf{U}^\top)(\mathbf{U} \mathbf{D}_t \mathbf{V}^\top) = \mathbf{U} \mathbf{D}_t^3 \mathbf{V}^\top$. Therefore

$$\begin{aligned} \mathbf{X}_{t+1} &= \frac{3}{2} \mathbf{U} \mathbf{D}_t \mathbf{V}^\top - \frac{1}{2} \mathbf{U} \mathbf{D}_t^3 \mathbf{V}^\top \\ &= \mathbf{U} \left(\frac{3}{2} \mathbf{D}_t - \frac{1}{2} \mathbf{D}_t^3 \right) \mathbf{V}^\top. \end{aligned}$$

272 Since $\mathbf{D}_t$ is diagonal,

$$\frac{3}{2} \mathbf{D}_t - \frac{1}{2} \mathbf{D}_t^3 = \text{diag}(f(p_t(s_1)), \dots, f(p_t(s_r))) = \text{diag}(p_{t+1}(s_1), \dots, p_{t+1}(s_r)).$$

273 Thus, the claim follows by induction. Since each $s_i = \sigma_i/\alpha$ belongs to $(0, 1]$, it follows by induction
274 and Lemma B.2 that for every $q \geq 0$,

$$0 \leq p_q(x) \leq 1 \quad \text{for all } x \in [0, 1],$$

275 and p_q is increasing on $[0, 1]$. Then from the above result, we have

$$\|T_q(\mathbf{M})\|_{\text{op}} = \max_{1 \leq i \leq r} p_q(s_i) \leq 1.$$

276 Therefore, the second assumption holds with $\rho_q(\mathbf{M}) = 0$. Now for $\delta_q(\mathbf{M})$ we compute

$$\begin{aligned} \langle \mathbf{M}, T_q(\mathbf{M}) \rangle &= \text{tr}(\mathbf{M}^\top T_q(\mathbf{M})) \\ &= \text{tr}((\mathbf{V} \Sigma \mathbf{U}^\top)(\mathbf{U} \mathbf{D}_q \mathbf{V}^\top)) \\ &= \text{tr}(\mathbf{V} \Sigma \mathbf{D}_q \mathbf{V}^\top) \\ &= \text{tr}(\Sigma \mathbf{D}_q) \\ &= \sum_{i=1}^r \sigma_i p_q(s_i) \\ &= \left(\frac{\sum_{i=1}^r \sigma_i p_q(s_i)}{\sum_{i=1}^r \sigma_i} \right) \|\mathbf{M}\|_*. \end{aligned}$$

277 Hence, the first assumption holds with

$$\delta_q(\mathbf{M}) = 1 - \frac{\sum_{i=1}^r \sigma_i p_q(s_i)}{\sum_{i=1}^r \sigma_i}.$$

278 This concludes the proof of the Theorem. □

Lemma B.2. For $f(x) := \frac{3}{2}x - \frac{1}{2}x^3$, we have

$$0 \leq f(x) \leq 1, \quad f(x) \geq x, \quad f'(x) \geq 0.$$

for every $x \in [0, 1]$.

279 *Proof.* Note that

$$f(x) - x = \frac{1}{2}x(1 - x^2) \geq 0,$$

280 so $f(x) \geq x$. Again, we have

$$1 - f(x) = 1 - \frac{3}{2}x + \frac{1}{2}x^3 = \frac{1}{2}(1 - x)^2(2 + x) \geq 0,$$

281 so $f(x) \leq 1$. Finally,

$$f'(x) = \frac{3}{2}(1 - x^2) \geq 0 \quad \text{for } x \in [0, 1].$$

282 □

283 **Step 5: a simpler admissible parameter $\bar{\delta}_q(M)$.** Because p_q is increasing on $[0, 1]$ and $s_1 \geq \dots \geq$
 284 s_r , we have

$$p_q(s_i) \geq p_q(s_r) \quad \text{for every } i = 1, \dots, r.$$

285 Therefore

$$\langle M, T_q(M) \rangle = \sum_{i=1}^r \sigma_i p_q(s_i) \geq p_q(s_r) \sum_{i=1}^r \sigma_i = p_q(s_r) \|M\|_*.$$

286 Thus (A1) also holds with

$$\bar{\delta}_q(M) := 1 - p_q(s_r).$$

287 **Step 6: a nonrecursive closed-form upper bound on $\bar{\delta}_q(M)$.** Define, for $x \in [0, 1]$,

$$e_t(x) := 1 - p_t(x)^2.$$

288 Let $y := p_t(x) \in [0, 1]$. Then

$$p_{t+1}(x) = f(y) = \frac{3}{2}y - \frac{1}{2}y^3.$$

289 Hence

$$e_{t+1}(x) = 1 - f(y)^2 = 1 - \frac{y^2(3 - y^2)^2}{4}.$$

290 Now set $e := 1 - y^2$. Then $y^2 = 1 - e$ and $3 - y^2 = 2 + e$, so

$$\begin{aligned} e_{t+1}(x) &= 1 - \frac{(1 - e)(2 + e)^2}{4} \\ &= 1 - \frac{4 - 3e^2 - e^3}{4} \\ &= \frac{1}{4}e^2(3 + e). \end{aligned}$$

291 Since $e \in [0, 1]$, we have $3 + e \leq 4$, and therefore

$$e_{t+1}(x) \leq e_t(x)^2.$$

292 By induction,

$$e_q(x) \leq e_0(x)^{2^q} = (1 - x^2)^{2^q}.$$

293 Recalling $e_q(x) = 1 - p_q(x)^2$, we conclude that

$$p_q(x) \geq \sqrt{1 - (1 - x^2)^{2^q}}.$$

294 Applying this with $x = s_r$ gives

$$\bar{\delta}_q(M) = 1 - p_q(s_r) \leq 1 - \sqrt{1 - (1 - s_r^2)^{2^q}}.$$

295 Finally, for any $z \in [0, 1]$,

$$1 - \sqrt{1 - z} = \frac{z}{1 + \sqrt{1 - z}} \leq z,$$

296 so with $z = (1 - s_r^2)^{2^q}$,

$$\bar{\delta}_q(M) \leq (1 - s_r^2)^{2^q}.$$

297 This proves the stated closed-form bounds.

298 If $s_r \geq \ell$, then by monotonicity of the right-hand side in s_r ,

$$\bar{\delta}_q(M) \leq 1 - \sqrt{1 - (1 - \ell^2)^{2^q}} \leq (1 - \ell^2)^{2^q}.$$

299 Together with $\rho_q(\ell) = 0$, this proves the uniform version of (A1)–(A2).

300 Moreover, the following simpler admissible choice also works:

$$\bar{\delta}_q(M) := 1 - p_q(s_r), \quad \rho_q(M) := 0.$$

301 Finally, one has the explicit nonrecursive bound

$$\bar{\delta}_q(M) \leq 1 - \sqrt{1 - (1 - s_r^2)^{2^q}} \leq (1 - s_r^2)^{2^q}.$$

302 Hence, if for some $\ell \in (0, 1]$ one has

$$\frac{\sigma_{\min}^+(M)}{\alpha} = s_r \geq \ell,$$

303 then (A1)–(A2) hold with the closed-form uniform parameters

$$\delta_q(\ell) := 1 - \sqrt{1 - (1 - \ell^2)^{2^q}}, \quad \rho_q(\ell) := 0,$$

304 and in particular also with the looser but simpler choice

$$\delta_q(\ell) = (1 - \ell^2)^{2^q}, \quad \rho_q(\ell) = 0.$$

Remark B.3. The same argument yields a stronger operator-norm approximation statement:

$$\|T_q(M) - O(M)\|_{\text{op}} = \max_{1 \leq i \leq r} |p_q(s_i) - 1| = 1 - p_q(s_r) = \bar{\delta}_q(M).$$

Thus the Newton–Schulz iterate is not only an (A1)–(A2) oracle, but is also operator-norm close to the exact compact polar factor.

Remark B.4. Without a lower bound on $s_r = \sigma_{\min}^+(M)/\alpha$, there is no nontrivial matrix-independent bound $\delta_q < 1$ for fixed q : indeed $p_q(0) = 0$, so $p_q(x) \rightarrow 0$ as $x \downarrow 0$, and consequently $\bar{\delta}_q(M) = 1 - p_q(s_r)$ can be arbitrarily close to 1 when the smallest positive singular value is tiny relative to α .

305 **Randomized smaller-subspace Newton–Schulz.** Let $M \in \mathbb{R}^{d \times d}$, let $Q \in \mathbb{R}^{d \times s}$ have orthonormal
 306 columns (possibly random), and define

$$P_Q := QQ^\top, \quad B := Q^\top M \in \mathbb{R}^{s \times d}.$$

307 Fix a normalization parameter $\alpha \geq \|B\|_{\text{op}}$, and define the rectangular Newton–Schulz iterates

$$Z_0 := \frac{B}{\alpha}, \quad Z_{t+1} := \frac{3}{2}Z_t - \frac{1}{2}Z_t Z_t^\top Z_t, \quad t = 0, 1, \dots, q-1.$$

308 The randomized smaller-subspace Newton–Schulz output is

$$T_{Q,q}(M) := QZ_q.$$

309 When Q is random, all constants below are random variables; the statements hold deterministically
 310 conditional on the realized Q .

311 **Theorem B.5** (Compressed Newton–Schulz satisfies (A1)–(A2)). Let $B = Q^\top M$, and let

312 $B = \hat{U} \hat{\Sigma} \hat{V}^\top$, $\hat{\Sigma} = \text{Diag}(\hat{\sigma}_1, \dots, \hat{\sigma}_{\hat{r}})$, $\hat{\sigma}_1 \geq \dots \geq \hat{\sigma}_{\hat{r}} > 0$,
be the compact SVD of B , where $\hat{r} = \text{rank}(B)$. Define

$$\widehat{s}_i := \frac{\widehat{\sigma}_i}{\alpha} \in (0, 1], \quad i = 1, \dots, \widehat{r},$$

313 and the scalar Newton–Schulz polynomials

$$f(x) := \frac{3}{2}x - \frac{1}{2}x^3, \quad p_0(x) := x, \quad p_{t+1}(x) := f(p_t(x)).$$

314

Then:

1. The compressed iterate has the singular-value form

$$Z_q = \widehat{U} \text{Diag}(p_q(\widehat{s}_1), \dots, p_q(\widehat{s}_{\widehat{r}})) \widehat{V}^\top,$$

and therefore

$$T_{Q,q}(M) = Q \widehat{U} \text{Diag}(p_q(\widehat{s}_1), \dots, p_q(\widehat{s}_{\widehat{r}})) \widehat{V}^\top.$$

2. The operator-norm assumption (A2) holds with the exact parameter

$$\rho_{Q,q}^*(M) = 0.$$

3. The inner-product assumption (A1) holds with the exact parameter

$$\delta_{Q,q}^*(M) := 1 - \frac{\sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i p_q(\widehat{s}_i)}{\|M\|_*}.$$

Equivalently,

$$\langle M, T_{Q,q}(M) \rangle = (1 - \delta_{Q,q}^*(M)) \|M\|_*.$$

4. Define the exact subspace-loss fraction

$$\xi_Q^*(M) := 1 - \frac{\|P_Q M\|_*}{\|M\|_*},$$

and the exact inner Newton–Schulz loss on the compressed matrix

$$\delta_q^{\text{NS}}(B) := 1 - \frac{\sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i p_q(\widehat{s}_i)}{\sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i}.$$

Then the exact parameter factorizes as

$$\delta_{Q,q}^*(M) = 1 - (1 - \xi_Q^*(M))(1 - \delta_q^{\text{NS}}(B)) = \xi_Q^*(M) + \delta_q^{\text{NS}}(B) - \xi_Q^*(M)\delta_q^{\text{NS}}(B).$$

5. Since

$$\|P_Q M\|_* \geq \|M\|_* - \|(I - P_Q)M\|_*,$$

the simpler admissible subspace-loss fraction

$$\xi_Q(M) := \frac{\|(I - P_Q)M\|_*}{\|M\|_*}$$

yields the valid parameter choice

$$\delta_{Q,q}(M) := 1 - (1 - \xi_Q(M))(1 - \delta_q^{\text{NS}}(B)), \quad \rho_{Q,q}(M) := 0.$$

6. If $B \neq 0$ and

$$\ell_B := \frac{\sigma_{\min}^+(B)}{\alpha} = \frac{\widehat{\sigma}_{\widehat{r}}}{\alpha},$$

then

$$\delta_q^{\text{NS}}(B) \leq \bar{\delta}_q^{\text{NS}}(B) := 1 - p_q(\ell_B) \leq 1 - \sqrt{1 - (1 - \ell_B^2)^{2^q}} \leq (1 - \ell_B^2)^{2^q}.$$

Hence (A1)–(A2) also hold with the closed-form admissible constants

$$\rho_{Q,q}(M) = 0,$$

and

$$\delta_{Q,q}(M) \leq 1 - (1 - \xi_Q(M)) \sqrt{1 - (1 - \ell_B^2)^{2^q}},$$

or, more simply,

$$\delta_{Q,q}(M) \leq \xi_Q(M) + (1 - \xi_Q(M))(1 - \ell_B^2)^{2^q}.$$

317 *Proof.* If $B = 0$, then $Z_t = 0$ for all t , hence $T_{Q,q}(M) = 0$, so (A1)–(A2) hold trivially with
 318 $(\delta, \rho) = (1, 0)$. Thus assume $B \neq 0$.

319 **Step 1: Singular-vector preservation on the compressed matrix.** We claim that

$$Z_t = \widehat{U} D_t \widehat{V}^\top, \quad D_t := \text{Diag}(p_t(\widehat{s}_1), \dots, p_t(\widehat{s}_{\widehat{r}})),$$

320 for every $t \geq 0$.

321 For $t = 0$,

$$Z_0 = \frac{B}{\alpha} = \widehat{U} \text{Diag}\left(\frac{\widehat{\sigma}_1}{\alpha}, \dots, \frac{\widehat{\sigma}_{\widehat{r}}}{\alpha}\right) \widehat{V}^\top = \widehat{U} D_0 \widehat{V}^\top.$$

322 Assume the claim holds at time t . Then

$$Z_t Z_t^\top Z_t = (\widehat{U} D_t \widehat{V}^\top)(\widehat{V} D_t \widehat{U}^\top)(\widehat{U} D_t \widehat{V}^\top) = \widehat{U} D_t^3 \widehat{V}^\top.$$

323 Therefore

$$Z_{t+1} = \frac{3}{2} \widehat{U} D_t \widehat{V}^\top - \frac{1}{2} \widehat{U} D_t^3 \widehat{V}^\top = \widehat{U} \left(\frac{3}{2} D_t - \frac{1}{2} D_t^3 \right) \widehat{V}^\top = \widehat{U} D_{t+1} \widehat{V}^\top.$$

324 This proves the claim.

325 Consequently,

$$T_{Q,q}(M) = Q \widehat{U} \text{Diag}(p_q(\widehat{s}_1), \dots, p_q(\widehat{s}_{\widehat{r}})) \widehat{V}^\top.$$

326 **Step 2: Scalar properties of the Newton–Schulz polynomial.** For $x \in [0, 1]$,

$$f(x) - x = \frac{1}{2}x(1 - x^2) \geq 0,$$

327 so $f(x) \geq x$. Also,

$$1 - f(x) = 1 - \frac{3}{2}x + \frac{1}{2}x^3 = \frac{1}{2}(1 - x)^2(2 + x) \geq 0,$$

328 so $f(x) \leq 1$. Finally,

$$f'(x) = \frac{3}{2}(1 - x^2) \geq 0,$$

329 so f is increasing on $[0, 1]$.

330 Since $\alpha \geq \|B\|_{\text{op}} = \widehat{\sigma}_1$, each $\widehat{s}_i \in (0, 1]$. By induction,

$$0 \leq p_q(x) \leq 1 \quad \text{for all } x \in [0, 1], \quad p_q \text{ is increasing on } [0, 1].$$

331 **Step 3: Proof of (A2) and the exact formula for $\rho_{Q,q}^*(M)$.** Because $Q^\top Q = I_s$, left multiplication
 332 by Q preserves Euclidean norms on $\mathbb{R}^s$. Hence

$$\|T_{Q,q}(M)\|_{\text{op}} = \|Q Z_q\|_{\text{op}} = \sup_{\|x\|_2=1} \|Q Z_q x\|_2 = \sup_{\|x\|_2=1} \|Z_q x\|_2 = \|Z_q\|_{\text{op}}.$$

333 Using the representation from Step 1,

$$\|Z_q\|_{\text{op}} = \max_{1 \leq i \leq \widehat{r}} p_q(\widehat{s}_i) \leq 1$$

334 by Step 2. Therefore

$$\|T_{Q,q}(M)\|_{\text{op}} \leq 1.$$

335 Thus (A2) holds with $\rho = 0$, and since $\rho \geq 0$ is required, the smallest admissible value is

$$\rho_{Q,q}^*(M) = 0.$$

336 **Step 4: Exact formula for $\langle M, T_{Q,q}(M) \rangle$.** Using $B = Q^\top M$,

$$\langle M, T_{Q,q}(M) \rangle = \langle M, Q Z_q \rangle = \langle Q^\top M, Z_q \rangle = \langle B, Z_q \rangle.$$

337 Now substitute the SVD forms of B and Z_q :

$$\begin{aligned}
\langle B, Z_q \rangle &= \text{tr}(B^\top Z_q) \\
&= \text{tr}\left((\widehat{V}\widehat{\Sigma}\widehat{U}^\top)(\widehat{U}D_q\widehat{V}^\top)\right) \\
&= \text{tr}(\widehat{V}\widehat{\Sigma}D_q\widehat{V}^\top) \\
&= \text{tr}(\widehat{\Sigma}D_q) \\
&= \sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i p_q(\widehat{s}_i).
\end{aligned}$$

338 Therefore

$$\langle M, T_{Q,q}(M) \rangle = \sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i p_q(\widehat{s}_i).$$

339 It follows that (A1) holds with equality for

$$\delta_{Q,q}^*(M) = 1 - \frac{\sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i p_q(\widehat{s}_i)}{\|M\|_*}.$$

340 This is the smallest admissible δ .

341 **Step 5: Exact factorization of the loss into a subspace term and an inner Newton–Schulz term.**

342 Define

$$\delta_q^{\text{NS}}(B) = 1 - \frac{\sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i p_q(\widehat{s}_i)}{\sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i}.$$

343 Then

$$\sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i p_q(\widehat{s}_i) = (1 - \delta_q^{\text{NS}}(B)) \sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i.$$

344 Also, since $P_Q M = Q(Q^\top M) = QB$, the nonzero singular values of $P_Q M$ equal those of B :

345 indeed, if $B = \widehat{U}\widehat{\Sigma}\widehat{V}^\top$, then

$$P_Q M = QB = (Q\widehat{U})\widehat{\Sigma}\widehat{V}^\top,$$

346 and $Q\widehat{U}$ has orthonormal columns because

$$(Q\widehat{U})^\top (Q\widehat{U}) = \widehat{U}^\top Q^\top Q \widehat{U} = I.$$

347 Hence

$$\|P_Q M\|_* = \|B\|_* = \sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i.$$

348 Therefore

$$\langle M, T_{Q,q}(M) \rangle = (1 - \delta_q^{\text{NS}}(B)) \|P_Q M\|_*.$$

349 Now define

$$\xi_Q^*(M) := 1 - \frac{\|P_Q M\|_*}{\|M\|_*}.$$

350 Then

$$\|P_Q M\|_* = (1 - \xi_Q^*(M)) \|M\|_*,$$

351 so

$$\langle M, T_{Q,q}(M) \rangle = (1 - \delta_q^{\text{NS}}(B)) (1 - \xi_Q^*(M)) \|M\|_*.$$

352 Comparing with the definition of $\delta_{Q,q}^*(M)$ yields

$$\delta_{Q,q}^*(M) = 1 - (1 - \xi_Q^*(M)) (1 - \delta_q^{\text{NS}}(B)),$$

353 which is the desired factorization.

354 **Step 6: A simpler admissible subspace term.** Since

$$M = P_Q M + (I - P_Q)M,$$

355 the triangle inequality for the nuclear norm gives

$$\|M\|_* \leq \|P_Q M\|_* + \|(I - P_Q)M\|_*.$$

356 Rearranging,

$$\|P_Q M\|_* \geq \|M\|_* - \|(I - P_Q)M\|_*.$$

357 After dividing by $\|M\|_*$, this becomes

$$1 - \xi_Q^*(M) = \frac{\|P_Q M\|_*}{\|M\|_*} \geq 1 - \frac{\|(I - P_Q)M\|_*}{\|M\|_*} = 1 - \xi_Q(M).$$

358 Therefore

$$\langle M, T_{Q,q}(M) \rangle = (1 - \delta_q^{\text{NS}}(B))(1 - \xi_Q^*(M))\|M\|_* \geq (1 - \delta_q^{\text{NS}}(B))(1 - \xi_Q(M))\|M\|_*.$$

359 Hence (A1) also holds with

$$\delta_{Q,q}(M) = 1 - (1 - \xi_Q(M))(1 - \delta_q^{\text{NS}}(B)).$$

360 **Step 7: Closed-form Newton–Schulz bound on the compressed matrix.** Let

$$e_t(x) := 1 - p_t(x)^2, \quad x \in [0, 1].$$

361 Set $y := p_t(x) \in [0, 1]$ and $e := 1 - y^2$. Then

$$p_{t+1}(x) = f(y) = \frac{3}{2}y - \frac{1}{2}y^3,$$

362 so

$$e_{t+1}(x) = 1 - f(y)^2.$$

363 Using $y^2 = 1 - e$, we compute

$$\begin{aligned} 1 - f(y)^2 &= 1 - \frac{y^2(3 - y^2)^2}{4} \\ &= 1 - \frac{(1 - e)(2 + e)^2}{4} \\ &= \frac{1}{4}e^2(3 + e) \leq e^2, \end{aligned}$$

364 because $0 \leq e \leq 1$ implies $3 + e \leq 4$. Thus

$$e_{t+1}(x) \leq e_t(x)^2.$$

365 Since $e_0(x) = 1 - x^2$, induction gives

$$e_q(x) \leq (1 - x^2)^{2^q}.$$

366 Therefore

$$p_q(x) \geq \sqrt{1 - (1 - x^2)^{2^q}}.$$

367 Now let $\ell_B = \widehat{\sigma}_{\widehat{r}}/\alpha$. Because p_q is increasing on $[0, 1]$,

$$p_q(\widehat{s}_i) \geq p_q(\ell_B) \quad \text{for all } i = 1, \dots, \widehat{r},$$

368 so

$$\delta_q^{\text{NS}}(B) = 1 - \frac{\sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i p_q(\widehat{s}_i)}{\sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i} \leq 1 - p_q(\ell_B).$$

369 Combining with the previous lower bound on $p_q(\ell_B)$ yields

$$\delta_q^{\text{NS}}(B) \leq 1 - \sqrt{1 - (1 - \ell_B^2)^{2^q}}.$$

370 Finally, for $z \in [0, 1]$,

$$1 - \sqrt{1 - z} = \frac{z}{1 + \sqrt{1 - z}} \leq z,$$

371 hence

$$\delta_q^{\text{NS}}(B) \leq (1 - \ell_B^2)^{2^q}.$$

372 Substituting either of these bounds into Step 6 gives the displayed closed-form bounds for $\delta_{Q,q}(M)$.

373 $\square$

Corollary B.6 (Explicit constants for randomized subspace iteration). Assume Q is produced by the randomized subspace iteration

$$Y = (MM^\top)^h M\Omega, \quad Q = \text{orth}(Y),$$

where $\Omega \in \mathbb{R}^{d \times (k+\rho)}$. Write the SVD of M as

$$M = [U_k \quad U_\perp] \begin{bmatrix} \Sigma_k & 0 \\ 0 & \Sigma_\perp \end{bmatrix} [V_k \quad V_\perp]^\top,$$

define

$$\gamma_k := \frac{\sigma_{k+1}(M)}{\sigma_k(M)}, \quad \Omega_1 := V_k^\top \Omega, \quad \Omega_2 := V_\perp^\top \Omega,$$

and assume that Ω_1 has full row rank. Then the randomized smaller-subspace Newton–Schulz output $T_{Q,q}(M)$ satisfies (A1)–(A2) with

$$\rho_{Q,q}(M) = 0,$$

and

$$\delta_{Q,q}(M) \leq 1 - \left(1 - \frac{\|\Sigma_\perp\|_* + \gamma_k^{2h} \|\Sigma_\perp \Omega_2 \Omega_1^\dagger\|_*}{\|M\|_*} \right) (1 - \delta_q^{\text{NS}}(Q^\top M)),$$

where

$$\delta_q^{\text{NS}}(Q^\top M) = 1 - \frac{\sum_{i=1}^{\hat{r}} \hat{\sigma}_i p_q(\hat{\sigma}_i/\alpha)}{\sum_{i=1}^{\hat{r}} \hat{\sigma}_i}.$$

If, in addition,

$$\frac{\sigma_{\min}^+(Q^\top M)}{\alpha} \geq \ell,$$

then

$$\delta_{Q,q}(M) \leq 1 - \left(1 - \frac{\|\Sigma_\perp\|_* + \gamma_k^{2h} \|\Sigma_\perp \Omega_2 \Omega_1^\dagger\|_*}{\|M\|_*} \right) \sqrt{1 - (1 - \ell^2)^{2q}}.$$

374 *Proof.* By the previous theorem, it is enough to upper bound

$$\xi_Q(M) = \frac{\|(I - P_Q)M\|_*}{\|M\|_*}.$$

375 For randomized subspace iteration, we assume the structural bound

$$\|(I - P_Q)M\|_* \leq \|\Sigma_\perp\|_* + \gamma_k^{2h} \|\Sigma_\perp \Omega_2 \Omega_1^\dagger\|_*.$$

376 Dividing by $\|M\|_*$ yields the displayed bound on $\xi_Q(M)$. The claimed formula for $\delta_{Q,q}(M)$ now
377 follows immediately from

$$\delta_{Q,q}(M) = 1 - (1 - \xi_Q(M))(1 - \delta_q^{\text{NS}}(Q^\top M)),$$

378 and the final inequality follows by substituting the closed-form Newton–Schulz bound

$$\delta_q^{\text{NS}}(Q^\top M) \leq 1 - \sqrt{1 - (1 - \ell^2)^{2q}}.$$

379

□

380 **Randomized compressed inexact polar decomposition.** Let $M \in \mathbb{R}^{d \times d}$, let $Q \in \mathbb{R}^{d \times s}$ have
 381 orthonormal columns (possibly random), and define

$$P_Q := QQ^\top, \quad B := Q^\top M \in \mathbb{R}^{s \times d}.$$

382 Let $S(B) \in \mathbb{R}^{s \times d}$ be any approximation to the compact polar factor $O(B)$. We lift it back to $\mathbb{R}^{d \times d}$
 383 via

$$T(M) := Q S(B) = Q S(Q^\top M).$$

384 We study whether $T(M)$ satisfies

$$\langle M, T(M) \rangle \geq (1 - \delta(M)) \|M\|_*, \quad (\text{A1})$$

$$\|T(M)\|_{\text{op}} \leq 1 + \rho(M). \quad (\text{A2})$$

385 **Auxiliary quantities.** Assume $M \neq 0$. Define the exact subspace-retention factor

$$\Gamma_Q(M) := \frac{\|P_Q M\|_*}{\|M\|_*} \in [0, 1],$$

386 and the exact subspace-loss fraction

$$\xi_Q^*(M) := 1 - \Gamma_Q(M) = 1 - \frac{\|P_Q M\|_*}{\|M\|_*}.$$

387 Also define the compressed progress factor

$$\Theta(B, S) := \begin{cases} \frac{\langle B, S(B) \rangle}{\|B\|_*}, & B \neq 0, \\ 0, & B = 0. \end{cases}$$

388 Finally, define the exact compressed norm-inflation parameter

$$\rho_B^*(S) := [\|S(B)\|_{\text{op}} - 1]_+.$$

389 **Theorem B.7** (Lifting an inexact compressed polar solver). Let $M \neq 0$, $Q^\top Q = I$, $B = Q^\top M$,

and $T(M) = Q S(B)$. Then:

(i) The exact identity

$$\langle M, T(M) \rangle = \Theta(B, S) \Gamma_Q(M) \|M\|_*$$

holds.

(ii) One has

$$\|T(M)\|_{\text{op}} = \|S(B)\|_{\text{op}}.$$

(iii) Consequently, $T(M)$ satisfies (A1)–(A2) with the exact minimal admissible parameters

$$\delta^*(M; Q, S) = [1 - \Theta(B, S) \Gamma_Q(M)]_+ = \left[1 - \frac{\langle B, S(B) \rangle}{\|M\|_*}\right]_+,$$

and

$$\rho^*(M; Q, S) = [\|S(B)\|_{\text{op}} - 1]_+.$$

(iv) If, moreover, the compressed solver $S(B)$ satisfies

$$\langle B, S(B) \rangle \geq (1 - \delta_B) \|B\|_*, \quad (\text{B1})$$

$$\|S(B)\|_{\text{op}} \leq 1 + \rho_B, \quad (\text{B2})$$

for some $\delta_B \in [0, 1]$, $\rho_B \geq 0$, then $T(M)$ satisfies (A1)–(A2) with

$$\delta(M) = 1 - (1 - \xi_Q(M))(1 - \delta_B), \quad \rho(M) = \rho_B,$$

for any $\xi_Q(M) \in [0, 1]$ such that

$$\|P_Q M\|_* \geq (1 - \xi_Q(M)) \|M\|_*.$$

In particular, the choice

$$\xi_Q(M) := \frac{\|(I - P_Q)M\|_*}{\|M\|_*}$$

is always valid.

390

391 *Proof.* If $B = 0$, then $T(M) = Q S(0)$. In the applications below we will have $S(0) = 0$, and then
392 (A1)–(A2) are trivial. So assume $B \neq 0$.

393 **Step 1: proof of the inner-product identity.** Using $B = Q^\top M$,

$$\langle M, T(M) \rangle = \langle M, Q S(B) \rangle = \langle Q^\top M, S(B) \rangle = \langle B, S(B) \rangle.$$

394 By the definition of $\Theta(B, S)$,

$$\langle B, S(B) \rangle = \Theta(B, S) \|B\|_*.$$

395 Thus

$$\langle M, T(M) \rangle = \Theta(B, S) \|B\|_*.$$

396 It remains to show that $\|B\|_* = \|P_Q M\|_*$. Let the compact SVD of B be

$$B = \hat{U} \hat{\Sigma} \hat{V}^\top.$$

397 Then

$$P_Q M = Q Q^\top M = Q B = (Q \hat{U}) \hat{\Sigma} \hat{V}^\top.$$

398 Since $Q^\top Q = I$ and $\hat{U}^\top \hat{U} = I$,

$$(Q \hat{U})^\top (Q \hat{U}) = \hat{U}^\top Q^\top Q \hat{U} = I.$$

399 Hence $Q \hat{U}$ has orthonormal columns, so the nonzero singular values of $P_Q M$ equal those of B .

400 Therefore

$$\|B\|_* = \|P_Q M\|_*.$$

401 Substituting into the previous identity,

$$\langle M, T(M) \rangle = \Theta(B, S) \|P_Q M\|_* = \Theta(B, S) \Gamma_Q(M) \|M\|_*.$$

402 This proves (i).

403 **Step 2: proof of the operator-norm identity.** Because $Q^\top Q = I$,

$$\|Qx\|_2 = \|x\|_2 \quad \text{for all } x \in \mathbb{R}^s.$$

404 Hence

$$\begin{aligned} \|T(M)\|_{\text{op}} &= \|Q S(B)\|_{\text{op}} \\ &= \sup_{\|x\|_2=1} \|Q S(B)x\|_2 \\ &= \sup_{\|x\|_2=1} \|S(B)x\|_2 \\ &= \|S(B)\|_{\text{op}}. \end{aligned}$$

405 This proves (ii).

406 **Step 3: exact parameters.** By Step 1,

$$\langle M, T(M) \rangle = \Theta(B, S) \Gamma_Q(M) \|M\|_*.$$

407 Therefore the smallest $\delta \in [0, 1]$ such that

$$\langle M, T(M) \rangle \geq (1 - \delta) \|M\|_*$$

408 is

$$\delta^*(M; Q, S) = [1 - \Theta(B, S) \Gamma_Q(M)]_+.$$

409 Using $\langle M, T(M) \rangle = \langle B, S(B) \rangle$, this can also be written as

$$\delta^*(M; Q, S) = \left[1 - \frac{\langle B, S(B) \rangle}{\|M\|_*} \right]_+.$$

410 Likewise, by Step 2, the smallest $\rho \geq 0$ such that

$$\|T(M)\|_{\text{op}} \leq 1 + \rho$$

411 is

$$\rho^*(M; Q, S) = [\|S(B)\|_{\text{op}} - 1]_+.$$

412 This proves (iii).

413 **Step 4: admissible parameters from a compressed (A1)–(A2) bound.** Assume (B1)–(B2). Then

$$\langle M, T(M) \rangle = \langle B, S(B) \rangle \geq (1 - \delta_B) \|B\|_* = (1 - \delta_B) \|P_Q M\|_*.$$

414 If $\|P_Q M\|_* \geq (1 - \xi_Q(M)) \|M\|_*$, then

$$\langle M, T(M) \rangle \geq (1 - \delta_B)(1 - \xi_Q(M)) \|M\|_*.$$

415 Thus (A1) holds with

$$\delta(M) = 1 - (1 - \xi_Q(M))(1 - \delta_B).$$

416 Also, by Step 2 and (B2),

$$\|T(M)\|_{\text{op}} = \|S(B)\|_{\text{op}} \leq 1 + \rho_B,$$

417 so (A2) holds with $\rho(M) = \rho_B$.

418 Finally, the choice

$$\xi_Q(M) := \frac{\|(I - P_Q)M\|_*}{\|M\|_*}$$

419 is valid because

$$M = P_Q M + (I - P_Q)M$$

420 and the triangle inequality for the nuclear norm gives

$$\|M\|_* \leq \|P_Q M\|_* + \|(I - P_Q)M\|_*.$$

421 Rearranging,

$$\|P_Q M\|_* \geq \|M\|_* - \|(I - P_Q)M\|_* = (1 - \xi_Q(M)) \|M\|_*.$$

422 This proves (iv). □

423 **Corollary: randomized subspace + cubic Newton–Schulz on B .** Let

$$B = Q^\top M, \quad Z_0 = \frac{B}{\alpha_B}, \quad Z_{t+1} = \frac{3}{2}Z_t - \frac{1}{2}Z_t Z_t^\top Z_t, \quad S_q^{(3)}(B) := Z_q,$$

424 where $\alpha_B \geq \|B\|_{\text{op}}$. Let the compact SVD of B be

$$B = \widehat{U} \widehat{\Sigma} \widehat{V}^\top, \quad \widehat{\Sigma} = \text{Diag}(\widehat{\sigma}_1, \dots, \widehat{\sigma}_{\widehat{r}}),$$

425 and define

$$\widehat{s}_i := \frac{\widehat{\sigma}_i}{\alpha_B} \in (0, 1], \quad p_0(x) := x, \quad p_{t+1}(x) := \frac{3}{2}p_t(x) - \frac{1}{2}p_t(x)^3.$$

426 Then

$$S_q^{(3)}(B) = \widehat{U} \text{Diag}(p_q(\widehat{s}_1), \dots, p_q(\widehat{s}_{\widehat{r}})) \widehat{V}^\top.$$

427 In particular,

$$\langle B, S_q^{(3)}(B) \rangle = \sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i p_q(\widehat{s}_i), \quad \|S_q^{(3)}(B)\|_{\text{op}} \leq 1.$$

428 Hence

$$T_{Q,q}^{(3)}(M) := Q S_q^{(3)}(Q^\top M)$$

429 satisfies (A1)–(A2) with the exact minimal parameters

$$\delta_{Q,q}^{(3),*}(M) = 1 - \frac{\sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i p_q(\widehat{s}_i)}{\|M\|_*}, \quad \rho_{Q,q}^{(3),*}(M) = 0.$$

430 Moreover, if

$$\ell_B := \frac{\sigma_{\min}^+(B)}{\alpha_B} = \frac{\widehat{\sigma}_{\widehat{r}}}{\alpha_B},$$

431 then the deterministic cubic Newton–Schulz bound gives

$$\langle B, S_q^{(3)}(B) \rangle \geq \sqrt{1 - (1 - \ell_B^2)^{2q}} \|B\|_*,$$

432 and therefore $T_{Q,q}^{(3)}(M)$ satisfies (A1)–(A2) with

$$\delta_{Q,q}^{(3)}(M) = 1 - (1 - \xi_Q(M)) \sqrt{1 - (1 - \ell_B^2)^{2q}}, \quad \rho_{Q,q}^{(3)}(M) = 0.$$

433 Using $1 - \sqrt{1 - z} \leq z$ for $z \in [0, 1]$, one also has the simpler bound

$$\delta_{Q,q}^{(3)}(M) \leq \xi_Q(M) + (1 - \xi_Q(M))(1 - \ell_B^2)^{2q}.$$

434 *Proof.* Apply the previous theorem with $S(B) = S_q^{(3)}(B)$. The quoted identities for $S_q^{(3)}(B)$,
 435 $\langle B, S_q^{(3)}(B) \rangle$, and $\|S_q^{(3)}(B)\|_{\text{op}}$ are exactly the deterministic cubic Newton–Schulz result applied to
 436 the compressed matrix B . $\square$

437 **Corollary: randomized subspace + quintic Newton–Schulz on B .** Let

$$B = Q^\top M, \quad Z_0 = \frac{B}{\alpha_B}, \quad Z_{t+1} = 2Z_t - \frac{3}{2}Z_t Z_t^\top Z_t + \frac{1}{2}(Z_t Z_t^\top)^2 Z_t, \quad S_q^{(5)}(B) := Z_q,$$

438 where $\alpha_B \geq \|B\|_{\text{op}}$. Let

$$p_0(x) := x, \quad p_{t+1}(x) := 2p_t(x) - \frac{3}{2}p_t(x)^3 + \frac{1}{2}p_t(x)^5.$$

439 If

$$B = \widehat{U} \widehat{\Sigma} \widehat{V}^\top, \quad \widehat{s}_i = \frac{\widehat{\sigma}_i}{\alpha_B},$$

440 then

$$S_q^{(5)}(B) = \widehat{U} \text{Diag}(p_q(\widehat{s}_1), \dots, p_q(\widehat{s}_{\widehat{r}})) \widehat{V}^\top.$$

441 Define

$$\ell_B := \frac{\sigma_{\min}^+(B)}{\alpha_B}.$$

442 Then the deterministic quintic Newton–Schulz result gives

$$\|S_q^{(5)}(B) - O(B)\|_{\text{op}} \leq \varepsilon_B, \quad \varepsilon_B := (1 - \ell_B)^{2^q}.$$

443 Consequently,

$$\langle B, S_q^{(5)}(B) \rangle \geq (1 - \varepsilon_B) \|B\|_*, \quad \|S_q^{(5)}(B)\|_{\text{op}} \leq 1 + \varepsilon_B.$$

444 Hence

$$T_{Q,q}^{(5)}(M) := Q S_q^{(5)}(Q^\top M)$$

445 satisfies (A1)–(A2) with

$$\delta_{Q,q}^{(5)}(M) = 1 - (1 - \xi_Q(M))(1 - \varepsilon_B) = \xi_Q(M) + (1 - \xi_Q(M))\varepsilon_B,$$

446 and

$$\rho_{Q,q}^{(5)}(M) = \varepsilon_B.$$

447 The exact minimal admissible parameters are

$$\delta_{Q,q}^{(5),*}(M) = \left[1 - \frac{\sum_{i=1}^{\hat{r}} \hat{\sigma}_i p_q(\hat{s}_i)}{\|M\|_*} \right]_+, \quad \rho_{Q,q}^{(5),*}(M) = \left[\max_{1 \leq i \leq \hat{r}} p_q(\hat{s}_i) - 1 \right]_+.$$

448 *Proof.* Apply the lifting theorem with $S(B) = S_q^{(5)}(B)$, and use the deterministic quintic Newton–
449 Schulz estimate on the compressed matrix B . From

$$\|S_q^{(5)}(B) - O(B)\|_{\text{op}} \leq \varepsilon_B$$

450 we obtain

$$\langle B, S_q^{(5)}(B) \rangle = \langle B, O(B) \rangle + \langle B, S_q^{(5)}(B) - O(B) \rangle \geq \|B\|_* - \|B\|_* \varepsilon_B = (1 - \varepsilon_B) \|B\|_*,$$

451 and

$$\|S_q^{(5)}(B)\|_{\text{op}} \leq \|O(B)\|_{\text{op}} + \|S_q^{(5)}(B) - O(B)\|_{\text{op}} \leq 1 + \varepsilon_B.$$

452 Now invoke the lifting theorem. □

Remark B.8 (Conditional, high-probability, and expectation forms). All the statements above are deterministic *conditional on the realized subspace Q* . Since Q is random, the quantities $\xi_Q(M)$ and ℓ_B are random.

If your randomized subspace method guarantees that, on an event $\mathcal{E}$,

$$\|P_Q M\|_* \geq (1 - \bar{\xi}) \|M\|_*,$$

then on the same event the lifted solver satisfies (A1)–(A2) with $\xi_Q(M)$ replaced by $\bar{\xi}$. In particular:

- for cubic Newton–Schulz on B ,

$$\delta \leq 1 - (1 - \bar{\xi}) \sqrt{1 - (1 - \ell_B^2)^{2^q}}, \quad \rho = 0;$$

- for quintic Newton–Schulz on B ,

$$\delta \leq \bar{\xi} + (1 - \bar{\xi})(1 - \ell_B)^{2^q}, \quad \rho \leq (1 - \ell_B)^{2^q}.$$

453 **Gaussian randomized subspace iteration with inexact polar decomposition on the compressed**

454 **matrix.** Let $M \in \mathbb{R}^{d \times d}$ be fixed, and let

$$M = U \Sigma V^\top, \quad \Sigma = \text{Diag}(\sigma_1, \dots, \sigma_r), \quad \sigma_1 \geq \dots \geq \sigma_r > 0,$$

455 be its compact SVD. Fix integers $1 \leq k < r$, $p \geq 2$, and $h, q \geq 0$, and set

$$\ell := k + p.$$

456 Draw a Gaussian sketch

$$\Omega \sim \mathcal{N}(0, 1)^{d \times \ell},$$

457 form

$$Y := (MM^\top)^h M \Omega, \quad Q := \text{orth}(Y), \quad P_Q := QQ^\top, \quad B := Q^\top M \in \mathbb{R}^{\ell \times d}.$$

458 On the compressed matrix B , run q steps of the cubic Newton–Schulz iteration with spectral normal-

459 ization:

$$Z_0 := \frac{B}{\|B\|_{\text{op}}}, \quad Z_{t+1} := \frac{3}{2}Z_t - \frac{1}{2}Z_t Z_t^\top Z_t, \quad t = 0, 1, \dots, q-1.$$

460 Define the lifted randomized update

$$T_{h,q}(M; \Omega) := QZ_q.$$

461 **Expected inexact-polar assumptions.** Since the construction is randomized through Ω , the natural
462 deterministic version of the inexact-polar assumption is:

$$\mathbb{E}_\Omega[\langle M, T_{h,q}(M; \Omega) \rangle] \geq (1 - \bar{\delta}_{h,q}(M)) \|M\|_*, \quad (\text{EA1})$$

$$\|T_{h,q}(M; \Omega)\|_{\text{op}} \leq 1 + \bar{\rho}_{h,q}(M) \quad \text{almost surely.} \quad (\text{EA2})$$

463 Here the expectation is only over the Gaussian sketch Ω , with M fixed.

464 **Scalar Newton–Schulz polynomials.** Define

$$\phi_0(x) := x, \quad \phi_{t+1}(x) := \frac{3}{2}\phi_t(x) - \frac{1}{2}\phi_t(x)^3, \quad t \geq 0.$$

465 **Theorem B.9** (Gaussian randomized subspace + cubic Newton–Schulz satisfies (EA1)–(EA2)).

Let

$$\gamma_k := \frac{\sigma_{k+1}}{\sigma_k} \in [0, 1).$$

Then the randomized update $T_{h,q}(M; \Omega)$ satisfies the following.

(i) Almost surely,

$$\|T_{h,q}(M; \Omega)\|_{\text{op}} \leq 1.$$

Hence (EA2) holds with the exact deterministic constant

$$\bar{\rho}_{h,q}(M) = 0.$$

(ii) The expected progress bound

$$\mathbb{E}_\Omega[\langle M, T_{h,q}(M; \Omega) \rangle] \geq \frac{1}{\sigma_1} \left[\sum_{j=1}^k \sigma_j^2 - \frac{k}{p-1} \gamma_k^{4h} \sum_{j>k} \sigma_j^2 \right]_+$$

holds. Therefore (EA1) is satisfied with the fully explicit deterministic constant

$$\bar{\delta}_{h,q}(M) = \left[1 - \frac{\left[\sum_{j=1}^k \sigma_j^2 - \frac{k}{p-1} \gamma_k^{4h} \sum_{j>k} \sigma_j^2 \right]_+}{\sigma_1 \sum_{j=1}^r \sigma_j} \right]_+.$$

In particular, this constant contains no random quantity such as ζ_Q .

(iii) The exact optimal deterministic expectation constant for (EA1) is

$$\bar{\delta}_{h,q}^*(M) := \left[1 - \frac{1}{\|M\|_*} \mathbb{E}_\Omega \left(\sum_{i=1}^{\text{rank}(B)} \hat{\sigma}_i \phi_q \left(\frac{\hat{\sigma}_i}{\hat{\sigma}_1} \right) \right) \right]_+,$$

where

$$B = \hat{U} \hat{\Sigma} \hat{V}^\top, \quad \hat{\Sigma} = \text{Diag}(\hat{\sigma}_1, \dots, \hat{\sigma}_{\text{rank}(B)}),$$

is the compact SVD of B . The constant in part (ii) is a fully explicit closed-form admissible upper bound on $\bar{\delta}_{h,q}^*(M)$.

466

467 *Proof.* We divide the proof into several steps.

468 **Step 1: singular-value form of the Newton–Schulz iterate on B .** Condition on a realization of Ω ,
469 and assume first that $B \neq 0$. Let

$$B = \hat{U} \hat{\Sigma} \hat{V}^\top, \quad \hat{\Sigma} = \text{Diag}(\hat{\sigma}_1, \dots, \hat{\sigma}_{\hat{r}}), \quad \hat{\sigma}_1 \geq \dots \geq \hat{\sigma}_{\hat{r}} > 0,$$

470 be the compact SVD of B , where $\hat{r} = \text{rank}(B)$. Since

$$Z_0 = \frac{B}{\|B\|_{\text{op}}} = \hat{U} \text{Diag} \left(\frac{\hat{\sigma}_1}{\hat{\sigma}_1}, \dots, \frac{\hat{\sigma}_{\hat{r}}}{\hat{\sigma}_1} \right) \hat{V}^\top = \hat{U} \text{Diag} \left(1, \frac{\hat{\sigma}_2}{\hat{\sigma}_1}, \dots, \frac{\hat{\sigma}_{\hat{r}}}{\hat{\sigma}_1} \right) \hat{V}^\top,$$

471 the normalized singular values of Z_0 lie in $[0, 1]$.

472 We claim that for every $t \geq 0$,

$$Z_t = \hat{U} \text{Diag} \left(\phi_t \left(\frac{\hat{\sigma}_1}{\hat{\sigma}_1} \right), \dots, \phi_t \left(\frac{\hat{\sigma}_{\hat{r}}}{\hat{\sigma}_1} \right) \right) \hat{V}^\top.$$

473 This is true for $t = 0$. If it holds at time t , then

$$Z_t Z_t^\top Z_t = \hat{U} D_t \hat{V}^\top \hat{V} D_t \hat{U}^\top \hat{U} D_t \hat{V}^\top = \hat{U} D_t^3 \hat{V}^\top,$$

474 where

$$D_t = \text{Diag} \left(\phi_t \left(\frac{\hat{\sigma}_1}{\hat{\sigma}_1} \right), \dots, \phi_t \left(\frac{\hat{\sigma}_{\hat{r}}}{\hat{\sigma}_1} \right) \right).$$

475 Hence

$$Z_{t+1} = \frac{3}{2}Z_t - \frac{1}{2}Z_t Z_t^\top Z_t = \widehat{U} \left(\frac{3}{2}D_t - \frac{1}{2}D_t^3 \right) \widehat{V}^\top = \widehat{U} D_{t+1} \widehat{V}^\top,$$

476 which proves the claim by induction.

477 Thus

$$Z_q = \widehat{U} \operatorname{Diag} \left(\phi_q \left(\frac{\widehat{\sigma}_1}{\widehat{\sigma}_1} \right), \dots, \phi_q \left(\frac{\widehat{\sigma}_{\widehat{r}}}{\widehat{\sigma}_1} \right) \right) \widehat{V}^\top.$$

478 **Step 2: scalar properties of the cubic Newton–Schulz map.** Let

$$f(x) := \frac{3}{2}x - \frac{1}{2}x^3.$$

479 For $x \in [0, 1]$,

$$f(x) - x = \frac{1}{2}x(1 - x^2) \geq 0,$$

480 so $f(x) \geq x$. Also,

$$1 - f(x) = 1 - \frac{3}{2}x + \frac{1}{2}x^3 = \frac{1}{2}(1 - x)^2(2 + x) \geq 0,$$

481 so $f(x) \leq 1$. Finally,

$$f'(x) = \frac{3}{2}(1 - x^2) \geq 0 \quad \text{for } x \in [0, 1],$$

482 so f is increasing on $[0, 1]$.

483 Since $\phi_{t+1} = f \circ \phi_t$ and $\phi_0(x) = x$, it follows by induction that for every $t \geq 0$ and every $x \in [0, 1]$,

$$0 \leq \phi_t(x) \leq 1, \quad \phi_t(x) \geq x.$$

484 **Step 3: proof of (EA2).** Using the singular-value representation from Step 1 and the scalar bounds
485 from Step 2,

$$\|Z_q\|_{\text{op}} = \max_{1 \leq i \leq \widehat{r}} \phi_q \left(\frac{\widehat{\sigma}_i}{\widehat{\sigma}_1} \right) \leq 1.$$

486 Since $Q^\top Q = I$, left multiplication by Q preserves the operator norm:

$$\|T_{h,q}(M; \Omega)\|_{\text{op}} = \|QZ_q\|_{\text{op}} = \|Z_q\|_{\text{op}} \leq 1.$$

487 Thus (EA2) holds almost surely with

$$\bar{\rho}_{h,q}(M) = 0.$$

488 **Step 4: pathwise lower bound for the inner product.** Because $B = Q^\top M$,

$$\langle M, T_{h,q}(M; \Omega) \rangle = \langle M, QZ_q \rangle = \langle Q^\top M, Z_q \rangle = \langle B, Z_q \rangle.$$

489 Using the SVD forms of B and Z_q ,

$$\langle B, Z_q \rangle = \sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i \phi_q \left(\frac{\widehat{\sigma}_i}{\widehat{\sigma}_1} \right).$$

490 By Step 2, $\phi_q(x) \geq x$ on $[0, 1]$. Therefore,

$$\langle B, Z_q \rangle \geq \sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i \cdot \frac{\widehat{\sigma}_i}{\widehat{\sigma}_1} = \frac{\|B\|_F^2}{\widehat{\sigma}_1} = \frac{\|B\|_F^2}{\|B\|_{\text{op}}}.$$

491 Since $B = Q^\top M$ and $Q^\top$ is a contraction in operator norm,

$$\|B\|_{\text{op}} \leq \|M\|_{\text{op}} = \sigma_1.$$

492 Hence

$$\langle M, T_{h,q}(M; \Omega) \rangle = \langle B, Z_q \rangle \geq \frac{\|B\|_F^2}{\sigma_1}.$$

493 Because $\|B\|_F = \|Q^\top M\|_F = \|P_Q M\|_F$, we have the pathwise bound

$$\langle M, T_{h,q}(M; \Omega) \rangle \geq \frac{\|P_Q M\|_F^2}{\sigma_1}. \quad (1)$$

494 **Step 5: an explicit bound on $\mathbb{E}_\Omega \|P_Q M\|_F^2$.** Write the SVD of M in the block form

$$M = [U_1 \quad U_2] \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix} [V_1 \quad V_2]^\top,$$

495 where $\Sigma_1 = \text{Diag}(\sigma_1, \dots, \sigma_k)$ and $\Sigma_2 = \text{Diag}(\sigma_{k+1}, \dots, \sigma_r)$.

496 Define

$$\Omega_1 := V_1^\top \Omega \in \mathbb{R}^{k \times (k+p)}, \quad \Omega_2 := V_2^\top \Omega \in \mathbb{R}^{(r-k) \times (k+p)}.$$

497 Because Ω is standard Gaussian and the Gaussian law is rotationally invariant, Ω_1 and Ω_2 are
498 independent standard Gaussian matrices. Also, Ω_1 has full row rank almost surely.

499 Now invoke the Frobenius-norm case of the standard structural bound for randomized subspace
500 iteration:

$$\|(I - P_Q)M\|_F^2 \leq \|\Sigma_2\|_F^2 + \gamma_k^{4h} \|\Sigma_2 \Omega_2 \Omega_1^\dagger\|_F^2. \quad (2)$$

501 Taking conditional expectation with respect to Ω_2 , given Ω_1 , yields

$$\mathbb{E}_{\Omega_2} \left[\|\Sigma_2 \Omega_2 \Omega_1^\dagger\|_F^2 \mid \Omega_1 \right] = \|\Sigma_2\|_F^2 \|\Omega_1^\dagger\|_F^2. \quad (3)$$

502 Indeed, conditional on Ω_1 , the matrix $\Omega_1^\dagger$ is deterministic and Ω_2 is still standard Gaussian, so (3) is
503 just the exact Gaussian identity for the squared Frobenius norm of a scaled Gaussian matrix.

504 Taking expectation in Ω_1 and using the exact Gaussian pseudoinverse moment,

$$\mathbb{E}_\Omega \|\Omega_1^\dagger\|_F^2 = \frac{k}{p-1}, \quad (4)$$

505 we obtain

$$\mathbb{E}_\Omega \|\Sigma_2 \Omega_2 \Omega_1^\dagger\|_F^2 = \frac{k}{p-1} \|\Sigma_2\|_F^2. \quad (5)$$

506 Taking expectation in (2) and substituting (5) gives

$$\mathbb{E}_\Omega \|(I - P_Q)M\|_F^2 \leq \left(1 + \frac{k}{p-1} \gamma_k^{4h}\right) \|\Sigma_2\|_F^2. \quad (6)$$

507 Next, since P_Q is an orthogonal projector,

$$\|P_Q M\|_F^2 = \|M\|_F^2 - \|(I - P_Q)M\|_F^2.$$

508 Taking expectation and using (6),

$$\begin{aligned} \mathbb{E}_\Omega \|P_Q M\|_F^2 &= \|M\|_F^2 - \mathbb{E}_\Omega \|(I - P_Q)M\|_F^2 \\ &\geq \|M\|_F^2 - \left(1 + \frac{k}{p-1} \gamma_k^{4h}\right) \|\Sigma_2\|_F^2 \\ &= \sum_{j=1}^k \sigma_j^2 - \frac{k}{p-1} \gamma_k^{4h} \sum_{j>k} \sigma_j^2. \end{aligned}$$

509 Therefore,

$$\mathbb{E}_\Omega \|P_Q M\|_F^2 \geq \left[\sum_{j=1}^k \sigma_j^2 - \frac{k}{p-1} \gamma_k^{4h} \sum_{j>k} \sigma_j^2 \right]_+. \quad (7)$$

510 **Step 6: expected (EA1) with an explicit deterministic constant.** Taking expectation in (1) and
 511 using (7),

$$\mathbb{E}_\Omega[\langle M, T_{h,q}(M; \Omega) \rangle] \geq \frac{1}{\sigma_1} \left[\sum_{j=1}^k \sigma_j^2 - \frac{k}{p-1} \gamma_k^{4h} \sum_{j>k} \sigma_j^2 \right]_+.$$

512 Since

$$\|M\|_* = \sum_{j=1}^r \sigma_j,$$

513 this can be rewritten as

$$\mathbb{E}_\Omega[\langle M, T_{h,q}(M; \Omega) \rangle] \geq (1 - \bar{\delta}_{h,q}(M)) \|M\|_*,$$

514 with

$$\bar{\delta}_{h,q}(M) = \left[1 - \frac{\left[\sum_{j=1}^k \sigma_j^2 - \frac{k}{p-1} \gamma_k^{4h} \sum_{j>k} \sigma_j^2 \right]_+}{\sigma_1 \sum_{j=1}^r \sigma_j} \right]_+.$$

515 This proves (EA1).

516 **Step 7: exact optimal deterministic expectation constants.** The exact optimal deterministic
 517 expectation constant for (EA2) is clearly

$$\bar{\rho}_{h,q}^*(M) = 0,$$

518 because $\|T_{h,q}(M; \Omega)\|_{\text{op}} \leq 1$ almost surely.

519 For (EA1), let

$$B = \widehat{U} \widehat{\Sigma} \widehat{V}^\top, \quad \widehat{\Sigma} = \text{Diag}(\widehat{\sigma}_1, \dots, \widehat{\sigma}_{\widehat{r}}).$$

520 From Step 4 we have the exact identity

$$\langle M, T_{h,q}(M; \Omega) \rangle = \sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i \phi_q \left(\frac{\widehat{\sigma}_i}{\widehat{\sigma}_1} \right).$$

521 Hence the smallest deterministic $\bar{\delta} \in [0, 1]$ satisfying (EA1) is

$$\bar{\delta}_{h,q}^*(M) := \left[1 - \frac{1}{\|M\|_*} \mathbb{E}_\Omega \left(\sum_{i=1}^{\widehat{r}} \widehat{\sigma}_i \phi_q \left(\frac{\widehat{\sigma}_i}{\widehat{\sigma}_1} \right) \right) \right]_+.$$

522 The constant from Step 6 is a fully explicit closed-form upper bound on $\bar{\delta}_{h,q}^*(M)$. □

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